A Nonparametric Eigenvalue-Regularized Integrated Covariance Matrix Estimator Using High-Frequency Data for Portfolio Allocation

Clifford Lam*and Phoenix Feng†

Department of Statistics, London School of Economics and Political Science

Abstract

In portfolio allocation of a large pool of assets, the use of high frequency data allows the corresponding high-dimensional integrated covariance matrix estimator to be more adaptive to local volatility features, while sample size is significantly increased. To ameliorate the bias contributed from the extreme eigenvalues of the sample integrated covariance matrix when the dimension p of the matrix is large relative to the average daily sample size n, and the contamination by microstructure noise, various researchers attempted regularization with specific assumptions on the true matrix itself, like sparsity or factor structure, which can be restrictive at times. With non-synchronous trading and contamination of microstructure noise, we propose a nonparametrically eigenvalue-regularized integrated covariance matrix estimator (NERIVE) which does not assume specific structures for the underlying integrated covariance matrix. We show that NERIVE is almost surely positive definite, with extreme eigenvalues shrunk nonlinearly under the high dimensional framework $p/n \to c > 0$. We also prove that almost surely, the minimum variance optimal weight vector constructed using NERIVE has maximum exposure upper bound of order $p^{-1/2}$ and actual risk upper bound of order p^{-1}. The practical performance of NERIVE is illustrated by comparing to the usual two-scale realized covariance matrix as well as some other nonparametric alternatives using different simulation settings and a real data set.

Key words and phrases. High frequency data; Microstructure noise; Non-synchronous trading; Integrated covariance matrix; Portfolio allocation; Nonlinear shrinkage.

*Clifford Lam is Associate Professor, Department of Statistics, London School of Economics. Email: C.Lam2@lse.ac.uk

†Phoenix Feng is PhD student, Department of Statistics, London School of Economics. Email: H.Feng2@lse.ac.uk

1 Introduction

In modern day finance, the so called tick-by-tick data on the prices of financial assets are readily available together with huge volume of other financial data. Advanced computational power and efficient data storage facilities mean that these data are analyzed on a daily basis by various market makers and academic researchers. While the Markowitz portfolio theory (Markowitz, 1952) is originally proposed for a finite number of assets using inter-day price data, the now easily accessible intra-day high frequency price data for a large number assets gives rise to new possibilities for efficient portfolio allocation, on top of the apparent increase in sample size for returns and volatility matrix estimation.

Certainly, the associated challenges for using high frequency data have to be overcome at the same time. One main challenge comes from the well documented market microstructure noise in the recorded tick-by-tick price data (Aït-Sahalia et al., 2005, Asparouhova et al., 2013). Another challenge comes from the non-synchronous trading times when more than one assets are considered. In terms of integrated volatility estimation, Xiu (2010) suggested a maximum likelihood approach for consistent estimation under market microstructure noise. Aït-Sahalia et al. (2010) proposed a quasi-maximum likelihood approach for estimating the covariance between two assets, while Zhang (2011) proposed a two or multi-scale covariance estimator to remove the bias accumulated due to the microstructure noise in the usual realized covariance formula, while overcoming the non-synchronous trading times problem by using previous-tick times (see Section 2 also). Other attempts to overcome these two challenges together include Barndorff-Nielsen et al. (2011) and Griffin and Oomen (2011), to name but a few.

When there is more than one asset to manage, the integrated covariance matrix for the asset returns is an important input for risk management or portfolio allocation. A large number of assets requires an estimation of a large integrated covariance matrix. Even in the simplest case of independent and identically distributed random vectors, random matrix theory tells us that the sample covariance matrix will have severely biased extreme eigenvalues (see chapter 5.2 of Bai and Silverstein (2010) for instance). While high frequency data increases the sample size significantly, the problems of non-synchronized trading times across assets and contamination of market microstructure noise described in the previous paragraph make it even harder to obtain a satisfactory integrated covariance matrix when the number of assets has the same order as the intra-day sample size. To solve this, Wang and Zou (2010) assumes a sparsity condition and use thresholding to regularize a two or multi-scale covariance matrix estimator based on previous-tick times. Tao et al. (2011) uses this thresholded estimator to find a factor model structure for the daily dynamics of the integrated volatility matrix. With respect to portfolio allocation, Fan et al. (2012) proposes to regularize the portfolio weights by constraining the gross exposure of the portfolio directly. They obtained remarkable results by using the pairwise refresh method to calculate individual elements in the two-scale covariance matrix.

In this paper, we focus on the bias problem in the extreme eigenvalues of the two-scale covariance matrix, and propose to shrink them using a data splitting method for nonlinear shrinkage similar to Lam (2016). One of the main contribution from Lam (2016) is that it provides a possible way to perform nonlinear shrinkage on more than just independent and identically distributed random vectors, as in the case for the nonlinear shrinkage formula in Ledoit and Wolf (2012). Ultimately, we do not need to assume a particular structure of the underlying integrated covariance matrix, and can still show that regularization on the extreme eigenvalues is achieved in the two-scale covariance matrix, resulting in a positive definite integrated covariance matrix estimator. At the same time, using its inverse in the construction of the minimum variance portfolio induces a natural upper bound on the maximum exposure of the portfolio, which decays at a rate of $p^{-1/2}$, where p is the number of assets. This implies that as we have more assets to choose from, we shall not over-invest in one particular asset based on the minimum variance portfolio, which can be risky as market conditions change. As illustrated in our empirical results in Section 5, such a maximum exposure decay is not necessarily enjoyed by other methods, for instance the gross exposure constraint by Fan et al. (2012). Our method also ensures the actual risk of the portfolio decays at a rate of p^{-1}. These results highlight the usefulness of shrinkage of eigenvalues in the context of portfolio allocation, and the effectiveness of nonlinear shrinkage in portfolio allocation is fully demonstrated in Section 5.

The rest of the paper is organized as follows. Section 2 presents the notations and model for the high frequency data and introduce our way to perform nonlinear shrinkage on the two scale covariance matrix estimator. Asymptotic theories and detailed assumptions can be found in Section 3. Practical concerns and implementation can be found in Section 4, while all simulations and a thorough empirical study are presented in Section 5. We give the conclusion of the paper in Section 6, before all the proofs of the theorems in the paper in Section 7.

2 Framework and Methodology

Consider p assets with log price process $\mathbf{X}_t = (X_t^{(1)}, \ldots, X_t^{(p)})^{\mathrm{T}}$, which obeys the following diffusion process

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \boldsymbol{\sigma}_t d\mathbf{W}_t, \quad t \in [0, 1], \tag{2.1}$$

so that the time period is normalized to have length 1. Denote L the integer representing the number of partitions of the data, typically the number of days on which the data spans, so that

$$0 = \tau_0 < \tau_1 < \cdots < \tau_L = 1,$$

and $(\tau_{\ell-1}, \tau_\ell]$ represents the ℓth partition. The process $\mathbf{W}_t$ is a p-dimensional standard Brownian motion. The drift $\boldsymbol{\mu}_t \in \mathbb{R}^p$ and the volatility $\boldsymbol{\sigma}_t \in \mathbb{R}^{p\times p}$ are assumed to be continuous in t. For each time interval $[a, b] \subset [0, 1]$, the corresponding integrated covariance matrix is defined as

$$\boldsymbol{\Sigma}(a, b) = \int_a^b \boldsymbol{\sigma}_u \boldsymbol{\sigma}_u^{\mathrm{T}} du.$$

Let $\{v_s\}, 1 \leq s \leq nL$ be the set of all-refresh times for the log prices in $\mathbf{X}_t$, where $n(\ell)$ is the number of all-refresh times at partition ℓ, with $\ell = 1, \ldots, L$, and $n = L^{-1} \sum_{\ell=1}^{L} n(\ell)$ is the average number of all-refresh times in a partition. To recall, an all-refresh time v_s is the time when all assets have been traded at least once from the last all-refresh time v_{s-1}. Let $t_s^j \in (v_{s-1}, v_s]$ be the sth previous-tick time for the jth asset, which is the last trading time before or at v_s. For non-synchronous trading, $t_s^{j_1} \neq t_s^{j_2}$ for $j_1 \neq j_2$ in general. Also, high-frequency prices are typically contaminated by microstructure noise, so that at the all-refresh time v_s, we only observe

$$\mathbf{Y}(s) = \mathbf{X}(s) + \boldsymbol{\epsilon}(s),\ s = 1, \ldots, nL, \tag{2.2}$$

where $\mathbf{X}(s) = (X_{t_s^1}^{(1)}, \ldots, X_{t_s^p}^{(p)})^{\mathrm{T}}$, $\boldsymbol{\epsilon}(s) = (\epsilon_{t_s^1}^{(1)}, \ldots, \epsilon_{t_s^p}^{(p)})^{\mathrm{T}}$, with $\boldsymbol{\epsilon}(\cdot)$ independent of $\mathbf{X}(\cdot)$. Such independence is in line with Barndorff-Nielsen et al. (2011) and Fan et al. (2012), for instance. Each $\boldsymbol{\epsilon}(s)$ is independent of each other with mean $\mathbf{0}$ and covariance matrix $\boldsymbol{\Sigma}_\epsilon$. Detailed assumptions can be found in Section 3.

2.1 Two-Scale Covariance Estimator

Contamination of microstructure noise in high-frequency data means that the usual realized covariance is heavily biased. Hence in Zhang (2011) a two-scale covariance estimator (TSCV) is introduced to remove this bias. In this paper, we use a slightly modified multivariate version of the two-scale covariance estimator, also by Zhang (2011). For $\ell = 1, \ldots, L$, define

$$\begin{aligned}
\langle \widehat{Y^{(i)}, Y^{(j)}} \rangle_\ell &= [Y^{(i)}, Y^{(j)}]_\ell^{(K)} - \frac{|S^\ell(K)|_K}{|S^\ell(1)|}[Y^{(i)}, Y^{(j)}]_\ell^{(1)}, \text{ with} \\
[Y^{(i)}, Y^{(j)}]_\ell^{(K)} &= \frac{1}{K}\sum_{r \in S^\ell(K)} (Y^{(i)}_{t^i_r} - Y^{(i)}_{t^i_{r-K}})(Y^{(j)}_{t^j_r} - Y^{(j)}_{t^j_{r-K}}), \text{ and} \\
S^\ell(K) = \{r : t^i_r, t^i_{r-K} &\in (\tau_{\ell-1}, \tau_\ell] \text{ for all } i\}, \ |S^\ell(K)|_K = \frac{|S^\ell(K)| - K + 1}{K}.
\end{aligned} \tag{2.3}$$

The quantity $\langle \widehat{Y^{(i)}, Y^{(j)}} \rangle_\ell$ is the TSCV introduced in Zhang (2011) for the ith and the jth assets using $J = 1$ and all the all-refresh data points in the ℓth partition $(\tau_{\ell-1}, \tau_\ell]$. With this, and defining $\widehat{\langle X \rangle} = \langle \widehat{X, X} \rangle$, we define the TSCV for the partition $(\tau_{\ell-1}, \tau_\ell]$ to be

$$\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \frac{1}{4}\left(\langle \widehat{Y^{(i)} + Y^{(j)}} \rangle_\ell^{(K)} - \langle \widehat{Y^{(i)} - Y^{(j)}} \rangle_\ell^{(K)} \right)_{1 \le i,j \le p}. \tag{2.4}$$

We suppressed the dependence on K in the notation $\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$ and all related definitions in the next section. In section 3, we show that K works well at the order $n^{2/3}$, which is indeed the order of magnitude suggested in Zhang (2011).

2.2 Our Proposed Integrated Covariance Matrix Estimator

Although the two-scale covariance estimator in (2.4) removes the bias contributed from the microstructure noise, it does not solve the bias issue for the extreme eigenvalues when p is large such that $p/n \to c > 0$. In order to regularize the integrated covariance matrix in the time period $(\tau_{j-1}, \tau_j]$, $j = 1, \ldots, L$, we follow Lam (2016) and use a rotation-equivariant estimator $\mathbf{\Sigma}(\mathbf{D}) = \mathbf{P}_{-j}\mathbf{D}\mathbf{P}_{-j}$, where $\mathbf{D}$ is a diagonal matrix, and $\mathbf{P}_{-j}$ is orthogonal such that

$$\widetilde{\mathbf{\Sigma}}_{-j} = \mathbf{P}_{-j}\mathbf{D}_{-j}\mathbf{P}_{-j}^{\mathrm{T}}, \quad j = 1, \ldots, L, \quad \text{with } \widetilde{\mathbf{\Sigma}}_{-j} = \sum_{\ell \ne j} \widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell). \tag{2.5}$$

Same as Lam (2016), we want to find $\mathbf{D}$ to solve

$$\min_{\mathbf{D}} \left\| \mathbf{P}_{-j}\mathbf{D}\mathbf{P}_{-j}^{\mathrm{T}} - \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \right\|_F, \tag{2.6}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. We choose $\mathbf{P}_{-j}$ as the matrix of eigenvectors since similar to Lam (2016), $\mathbf{P}_{-j}$ is independent of $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$, so that ultimately regularization can be achieved through setting $\mathbf{D} = \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$ as in the integrated covariance matrix estimator (2.7).

In using $\mathbf{P}_{-j}$, we also assume that the integrated covariance matrix $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$ shares a very similar set of eigenvectors as $\mathbf{\Sigma}_{-j} = \sum_{\ell \neq j} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)$, such that $\mathbf{P}_{-j}$ is a reasonable estimator for the set of eigenvectors of $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$. One scenario for this to happen is that the integrated covariance matrix $\mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)$ for each partition $\ell = 1, \ldots, L$ are the same. This is especially true in practice if each partition $(\tau_{\ell-1}, \tau_\ell]$ represents a trading day, with data in terms of each trading day being weakly stationary. In this paper, we indeed set each $(\tau_{\ell-1}, \tau_\ell]$ to be a trading day, so that there are L days of data and nL data points in total.

With the above, we define our integrated covariance matrix estimator for the partition $(\tau_{j-1}, \tau_j]$ to be

$$\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) = \mathbf{P}_{-j} \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \mathbf{P}_{-j}^{\mathrm{T}}. \tag{2.7}$$

The overall integrated covariance matrix estimator for the period $[0, 1]$ is then defined to be

$$\widehat{\mathbf{\Sigma}}(0, 1) = \sum_{j=1}^{L} \widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) = \sum_{j=1}^{L} \mathbf{P}_{-j} \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \mathbf{P}_{-j}) \mathbf{P}_{-j}^{\mathrm{T}}, \tag{2.8}$$

The estimator in (2.7) is inspired by the sample splitting idea used in Abadir et al. (2014), where the $p \times n$ data $\mathbf{Y}$ is split into two independent partitions, say $\mathbf{Y} = (\mathbf{Y}_1, \mathbf{Y}_2)$ with $\mathbf{Y}_i$ of size $p \times n_i$. In Lam (2016), while eigen-decompositions are carried out for both partitions such that $\widetilde{\mathbf{\Sigma}}_i = n_i^{-1} \mathbf{Y}_i \mathbf{Y}_i^{\mathrm{T}} = \mathbf{P}_i \mathbf{D}_i \mathbf{P}_i^{\mathrm{T}}$, an estimator for the covariance matrix $\mathbf{\Sigma}$ is proposed as

$$\widehat{\mathbf{\Sigma}} = \mathbf{P}_1 \mathrm{diag}(\mathbf{P}_1^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}_2 \mathbf{P}_1) \mathbf{P}_1^{\mathrm{T}}.$$

Lam (2016) proved the asymptotic almost sure positive definiteness of this estimator, and showed almost sure regularization of the eigenvalues to $\mathrm{diag}(\mathbf{P}_1^{\mathrm{T}} \mathbf{\Sigma} \mathbf{P}_1)$, where $\mathbf{\Sigma}$ is the true covariance matrix. Analogous to the regularization coming from the independence of $\mathbf{P}_1$ and $\widetilde{\mathbf{\Sigma}}_2$ for the above estimator, our integrated covariance matrix estimator (2.7) has regularization comes from the independence of $\mathbf{P}_{-j}$ and $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$, as they are computed from asset returns excluding the jth partition, and those within the jth partition respectively, which are independent of each other from the diffusion model in (2.1) and the serial independence of microstructure noise assumed in (2.2).

3 Asymptotic Theory

In this section, we show that our proposed estimator (2.7) in the jth partition of the data is asymptotically close to the corresponding ideal rotation-equivariant estimator

$$\boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1},\tau_j) = \mathbf{P}_{-j}\text{diag}(\mathbf{P}_{-j}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{P}_{-j})\mathbf{P}_{-j}^{\mathrm{T}}. \tag{3.1}$$

We first introduce some assumptions for our theorems to hold. In the following and hereafter, we denote $\lambda_{\min}(\cdot)$ the minimum eigenvalue of a square matrix, and $a \asymp b$ means that $a = O(b)$ and $b = O(a)$.

(A1) The drift process $\{\boldsymbol{\mu}_t\}_{t\in[0,1]}$ in model (2.1) is such that $\boldsymbol{\mu}_t = \mathbf{0}$ for all $t \in [0,1]$.

(A2) The volatility process $\{\boldsymbol{\sigma}_u\}_{u\in[0,1]}$ satisfies

$$0 < C_1 \leq \min_{u\in[0,1]} \lambda_{\min}(\boldsymbol{\sigma}_u\boldsymbol{\sigma}_u^{\mathrm{T}}) \leq \max_{u\in[0,1]} \lambda_{\max}(\boldsymbol{\sigma}_u\boldsymbol{\sigma}_u^{\mathrm{T}}) \leq C_2 < \infty$$

uniformly as $p \to \infty$, where $C_1, C_2 > 0$ are generic constants.

(A3) The processes $\boldsymbol{\epsilon}(\cdot)$ and $\mathbf{X}(\cdot)$ defined in (2.2) are independent of each other. For fixed i, each $\epsilon_{t_s^i}^{(i)}$ is independent of each other for $s = 1, \ldots, nL$, and is normally distributed. The covariance matrix of $\boldsymbol{\epsilon}(s)$ is $\boldsymbol{\Sigma}_\epsilon$ for each s, which has uniformly bounded eigenvalues as $p \to \infty$.

(A4) The observation times are independent of $\mathbf{X}(\cdot)$ and $\boldsymbol{\epsilon}(\cdot)$, and the partition boundaries τ_ℓ, $\ell = 0, 1, \ldots, L$, satisfy $0 < C_3 \leq \min_{\ell=1,\ldots,L} L(\tau_\ell - \tau_{\ell-1}) \leq \max_{\ell=1,\ldots,L} L(\tau_\ell - \tau_{\ell-1}) \leq C_4 < \infty$, where C_3, C_4 are generic constant. Also, the all-refresh times v_s, $s = 1, \ldots, nL$ satisfy $\max_{s=1,\ldots,nL} nL(v_s - v_{s-1}) \leq C_5$ for a generic constant $C_5 > 0$. Moreover, $\max_{\ell=1,\ldots,L} L(\tau_\ell - v_{n(\ell)}) = o(1)$.

(A5) In each partition ℓ, the integrated covariance matrix $\boldsymbol{\Sigma}(\tau_{\ell-1}, \tau_\ell)$ has the same set of eigenvectors and order of eigenvalues.

(A6) For the TSCV parameters, we set $J = 1$ and $K \asymp n^{2/3}$, with $L = O(n^{1/13})$.

Assumption (A1) is for simplicity of our proofs. It can be extended to $\{\boldsymbol{\mu}_t\}$ being locally bounded because of the way $[Y^{(i)}, Y^{(j)}]_\ell^{(K)}$ is defined. The independence parts in Assumption (A3) is seen in Fan et al. (2012) and also Barndorff-Nielsen et al. (2011), but we relax their assumptions to allow for correlated microstructure noise components. The normality assumption for $\boldsymbol{\epsilon}(s)$ in (A3) can be replaced by higher order moment assumptions.

The first part of Assumption (A4) is automatically satisfied if the boundary set $\{\tau_\ell\}_{0\leq\ell\leq L}$ is pre-set, for instance, to be the daily opening or closing time of the L days of data. As described in Section 2.2, we indeed set τ_ℓ to be the closing time of the ℓth trading day, so that Assumption (A5) is satisfied in practice if the data, viewing in terms of each trading day, is weakly stationary. Assumptions (A2) and (A4) are there to control the size of the eigenvalues of $\mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)$ for each $\ell = 1, \ldots, L$.

In Assumption (A6), the parameter K in the TSCV is chosen to be exactly as what Zhang (2011) suggested. With this order of magnitude for K, we need L to be going to infinity with order as specified in (A6) in order to control the bias contributed from the microstructure noise. The order for K can actually be changed with an accompanying change in the order for L. We use $K \asymp n^{2/3}$ only just for the simplicity in the proof and ease of demonstration.

Theorem 1. *Let Assumptions (A1) to (A6) hold. For the all-refresh log-price data $\mathbf{Y}(s)$, $s = 1, \ldots, nL$ in (2.2), the integrated covariance matrix estimator constructed in (2.7) satisfies, as $n, p \to \infty$ such that $p/n \to c > 0$,*

$$\max_{j=1,\ldots,L} \left\| \widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p \right\| \stackrel{a.s.}{\to} 0,$$

where $\|\cdot\|$ denotes the L_2 norm of a matrix.

The proof can be found in Section 7. Since $\mathbf{P}_{-j}$ is orthogonal, it is easy to see that $\mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_j)$ in (3.1) has

$$\mathrm{Cond}(\mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_{j-1})) \leq \mathrm{Cond}(\mathbf{\Sigma}(\tau_{j-1}, \tau_j)),$$

where $\mathrm{Cond}(\cdot)$ is the condition number of a matrix, defined by dividing the maximum over the minimum magnitude of eigenvalue of the matrix. Theorem 1 then implies that almost surely, as $n, p \to \infty$ such that $p/n \to c > 0$,

$$\mathrm{Cond}(\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_{j-1})) \leq \mathrm{Cond}(\mathbf{\Sigma}(\tau_{j-1}, \tau_j)).$$

This is the result of nonlinear shrinkage of eigenvalues of $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$. Incidentally, since all eigenvalues of $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ are non-negative, this also proves the following.

Corollary 2. *Let Assumptions (A1) to (A6) hold. Then as $n, p \to \infty$ such that $p/n \to c > 0$, the integrated covariance matrix estimator $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ in (2.7), and also $\widehat{\mathbf{\Sigma}}(0, 1)$ in (2.8), are almost surely positive definite as long as $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$ and $\mathbf{\Sigma}(0, 1)$ are.*

This corollary shows that the positive definiteness of an integrated covariance matrix is preserved in our proposed estimator almost surely as we have large enough sample size. In practice, like NERCOME introduced in Lam (2016) (see also Section 5 for simulation results in portfolio analysis), we always have positive definiteness of the estimator with a moderate sample size n and a similar dimension p.

3.1 Application to Portfolio Allocation

Similar to Ledoit and Wolf (2012) and Lam (2016), instead of comparing the performance of $\widehat{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ with $\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)$, we compare it with the ideal rotation-equivariant estimator $\boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j)$ in (3.1), which replaces $\widetilde{\boldsymbol{\Sigma}}(\tau_{j-1}, \tau_j)$ in (2.7) by the true integrated covariance matrix $\boldsymbol{\Sigma}(\tau_{j-1}, \tau_j)$. In terms of the optimal portfolio weights for the minimum-variance portfolio, defining $\mathbf{1}_p$ as a column vector of p ones, we define

$$\widehat{\mathbf{w}}_{\text{opt}} = \frac{\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}, \quad \mathbf{w}_{\text{Ideal}} = \frac{\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p},$$

where

$$\boldsymbol{\Sigma}_{\text{Ideal}}(0,1) = \sum_{j=1}^{L} \boldsymbol{\Sigma}_{\text{Ideal}}(\tau_{j-1}, \tau_j).$$

Since both are minimum variance portfolios over the period $[0, 1]$, we naturally want to compare their associated risks, defined by

$$R(\mathbf{w}) = \mathbf{w}^{\mathrm{T}}\boldsymbol{\Sigma}(0,1)\mathbf{w}$$

for a particular portfolio weight vector $\mathbf{w}$. Define the efficiency of $\widehat{\mathbf{w}}_{\text{opt}}$ relative to $\mathbf{w}_{\text{Ideal}}$ by

$$\text{Eff}(\mathbf{w}_{\text{Ideal}}, \widehat{\mathbf{w}}_{\text{opt}}) = \frac{R(\mathbf{w}_{\text{Ideal}})}{R(\widehat{\mathbf{w}}_{\text{opt}})}. \tag{3.2}$$

With this definition, the efficiency is higher when $\widehat{\mathbf{w}}_{\text{opt}}$ has a smaller risk, and vice versa.

Theorem 3. *Let Assumptions (A1) to (A6) hold. Then as $n, p \to \infty$ such that $p/n \to c > 0$, we have* $\text{Eff}(\mathbf{w}_{\text{Ideal}}, \widehat{\mathbf{w}}_{\text{opt}}) \overset{a.s.}{\to} 1$.

The proof of this theorem is in Section 7. It says that the risk of the minimum variance portfolio constructed from using $\widehat{\boldsymbol{\Sigma}}(0,1)$ is asymptotically the same as that constructed using $\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)$ almost surely. The practical performance of $\widehat{\mathbf{w}}_{\text{opt}}$ is illustrated in the empirical analysis in Section 5.

3.1.1 Maximum exposure and actual risk upper bounds

Unlike Fan et al. (2012) which constraints the gross exposure $\|\mathbf{w}\|_1 = \sum_j |w_j|$ of a portfolio weight vector $\mathbf{w}$ explicitly through a tuning parameter, our method enjoys a natural upper bound on the maximum exposure asymptotically almost surely. The maximum exposure of a portfolio vector $\mathbf{w}$ is defined as $\|\mathbf{w}\|_{\max} = \max_i |w_i|$. At the same time, the actual risk $R(\widehat{\mathbf{w}}_{\text{opt}})$ also has a natural upper bound, as presented below.

Theorem 4. *Let Assumptions (A1) to (A6) hold. Then as $n, p \to \infty$ such that $p/n \to c > 0$, the maximum exposure and the actual risk of $\widehat{\mathbf{w}}_{opt}$ satisfy respectively,*

$$p^{1/2}\left\|\widehat{\mathbf{w}}_{\text{opt}}\right\|_{\max} \leq \text{Cond}(\mathbf{\Sigma}(0,1)), \qquad pR(\widehat{\mathbf{w}}_{\text{opt}}) \leq \text{Cond}^2(\mathbf{\Sigma}(0,1))\lambda_{\max}(\mathbf{\Sigma}(0,1))$$

almost surely, where $\lambda_{\max}(\cdot)$ denotes the maximum eigenvalue of a matrix.

The proof of this theorem is in Section 7. The gross exposure constraint by Fan et al. (2012) is useful in constraining the total exposure of a portfolio and obtaining special ones like the no-short-sale portfolio (by setting $\left\|\mathbf{w}\right\|_1 \leq 1$). In practice, as illustrated by our simulation experiments and real data analysis in Section 5, the maximum exposure can still be large while the gross exposure constraint is in place. This may not be ideal when there are many other assets to diversify the portfolio, especially when market conditions change such that a single stock in which the portfolio invests heavily changes in direction and thus results in a loss. Our method, on the other hand, has an almost sure upper bound which decays as p increases by Theorem 4. As illustrated in Section 5, the maximum exposure in $\widehat{\mathbf{w}}_{\text{opt}}$ is much smaller than other state-of-the-art methods, which can be important in a volatile market.

It is easy to see that the actual risk upper bound of $\widehat{\mathbf{w}}_{\text{opt}}$ has the same rate of decay as the equal weight portfolio $\mathbf{w}_{\text{equal}} = p^{-1}\mathbf{1}_p$, which has the upper bound being $p^{-1}\lambda_{\max}(\mathbf{\Sigma}(0,1))$. However, $\widehat{\mathbf{w}}_{\text{opt}}$ is certainly different from the equal weight portfolio since $\widehat{\mathbf{w}}_{\text{opt}}$ is minimizing the risk, as illustrated in Tables 4 and 5 in our empirical results in Section 5, with quite different maximum exposure shown.

4 Practical Implementation

There are two parameters that we can tune for potentially better performance, namely the partition $(\tau_{j-1}, \tau_j]$ of the period $[0,1]$ (thus also determining L itself which represents the number of partitions), and the scale parameter K used in the TSCV in (2.4). For example, if we are given a period of 10 days of tick-by-tick data, if we set $(\tau_{j-1}, \tau_j]$ to be one day, then $L = 10$. Similar to the function $g(m)$ in equation (4.7) of Lam (2016), we propose to minimize the following criterion for a good choice of $\boldsymbol{\tau} = \{\tau_j\}_{0 \leq j \leq L}$ and K:

$$g(\boldsymbol{\tau}, K) = \left\| \sum_{j=1}^{L} \left(\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) - \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \right) \right\|_F^2, \tag{4.1}$$

where $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ and $\widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ are defined in (2.4) and (2.7) respectively. In practice, for a given L, we can divide $[0,1]$ into equal partitions, so that $\boldsymbol{\tau}$ is then determined.

Recall that Assumption (A5) requires each integrated covariance matrix $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$ for $j = 1, \ldots, L$

shares the same set of eigenvectors. Since intraday volatility can change rapidly, having $(\tau_{j-1}, \tau_j]$ to be smaller than a trading day will likely to result in very different eigenvectors in different $\mathbf{\Sigma}(\tau_{j-1}, \tau_j)$. Hence we recommend keeping each trading day as a natural partition of the data, with L then being the number of trading days.

For the choice of K, since we are using $K \asymp n^{2/3}$ as in Zhang (2011), we can search $K = [bn^{2/3}]$ on a preset grid of constant b. In practice, we found from our simulation results and real data analysis that using $b = 1$ provide good results, and portfolio performance is not too different from using other values of b, hence in this paper we use $b = 1$.

5 Empirical Results

5.1 Simulation

In this section, we simulate high frequency trading transactions of 100 stocks for one year (250 trading days). Our method (NERIVE) is demonstrated and compared to different state-of-the-art methods, including the low frequency Nonparametric Eigenvalue-Regularized COvariance Matrix Estimator (NERCOME) by Lam (2016), the Two-Scale Covariance estimator (TSCV) by Zhang (2011), and the Gross Exposure Constraint (GEC) method based on TSCV by Fan et al. (2012).

Following Barndorff-Nielsen et al. (2011) and Fan et al. (2012), we simulate the price processes and the asynchronous transaction times independently. The observed log-price is defined as $X_t^{o(i)} = X_t^{(i)} + \varepsilon_t^{(i)}$, where $X_t^{(i)}$ represents the latent log-price, and the microstructure noise has $\varepsilon_t^{(i)} \overset{iid}{\sim} N(0, 0.0005^2)$. We generate $p = 100$ latent log-prices by the following multivariate factor model with stochastic volatilities:

$$dX_t^{(i)} = \mu^{(i)}dt + \rho^{(i)}\sigma_t^{(i)}dB_t^{(i)} + \sqrt{1 - (\rho^{(i)})^2}\sigma_t^{(i)}dW_t + C\nu^{(i)}dZ_t, \quad i = 1, ..., 100, \tag{5.1}$$

where $\{W_t\}$, $\{Z_t\}$ and the $\{B_t^{(i)}\}$'s are independent standard Brownian motions. The process $\{Z_t\}$ imitates a strong market factor. The constant $C = \mathbf{1}_{\{\text{model 2}\}}$ is 0 for the first model we consider, so that there are no pervasive market factors. For the second model, $C = 1$, so that it contains a pervasive market factor. The spot volatility $\sigma_t^{(i)} = exp(\varrho_t^{(i)})$ follows the independent Ornstein-Uhlenbeck process

$$d\varrho_t^{(i)} = \alpha^{(i)}(\beta_0^{(i)} - \varrho_t^{(i)})dt + \beta_1^{(i)}dU_t^{(i)},$$

where the $\{U_t^{(i)}\}$'s are independent standard Brownian motions. Other parameters of $X_t^{(i)}$ are set at $(\mu^{(i)}, \beta_0^{(i)}, \beta_1^{(i)}, \alpha^{(i)}, \rho^{(i)}) = (0.03x_1^{(i)}, -x_2^{(i)}, 0.75x_3^{(i)}, -1/40x_4^{(i)}, -0.7C)$ and $\nu^{(i)} = exp(\beta_0^{(i)})$, where $x_j^{(i)}$ is independent uniformly distributed on the interval $[0.7, 1.3]$. The initial value of each log-price is set at $X_0^{(i)} = 1$ and the starting spot volatility $\varrho_0^{(i)} = 0$. Setting $\rho^{(i)} = -0.7C$ means that for model 1 we do not have any factors, while for model 2 we have a pervasive factor and also other factors related to $\{W_t\}$.

For the transaction times, we generate 100 different Poisson processes with intensities $\lambda_1, ..., \lambda_{100}$ respectively. Since the normal trading time for one day is 23400 seconds, λ_i is set to be $0.01i \times 23400$, where $i = 1, ..., 100$.

5.2 Comparison of portfolio allocation performance

To compare the performance of different methods, we construct portfolio weights according to the minimum variance portfolio

$$\mathbf{w}_{\text{opt}} = \frac{\boldsymbol{\Sigma}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}(0,1)^{-1}\mathbf{1}_p}, \quad \text{which solves} \min_{\mathbf{w}:\mathbf{w}^{\mathrm{T}}\mathbf{1}_p=1} \mathbf{w}^{\mathrm{T}}\boldsymbol{\Sigma}(0,1)\mathbf{w}.$$

We first set the benchmark for comparisons. Following Fan et al. (2012), we create a theoretical portfolio $\mathbf{w}_{\text{theo}}$, which is a minimum variance portfolio with $\boldsymbol{\Sigma}(0,1)$ evaluated using the simulated latent log-prices at the finest grid (1 per second). For all other methods, we use the all-refresh time points evaluated from the data (we do not hold positions overnight for all methods to avoid overnight price jumps, since they are not what our study is about). We do not use the pairwise refresh method for the GEC to save significant computational time in both the simulations and later the empirical exercises, as well as that the features of our method can be compared more directly to those of GEC.

We invest 1 unit of capital to the different portfolios above at a certain start date (e.g., day 11 if we are using a 10-day training window) and rebalance the portfolio weights weekly or daily, moving the training window along one week or one day respectively. There are two investment strategies for comparisons under each model 1 or 2. The first one rebalances the portfolio weekly with a 10-week training window. The second one rebalances the portfolio daily with a 10-day training window. For each strategy, we compare $\mathbf{w}_{\text{theo}}$ with portfolio weight having $\boldsymbol{\Sigma}(0,1)$ in each investment horizon estimated by NERIVE, NERCOME (for the case with 10-week training window) and TSCV. We also compare them with 5 GEC portfolios with constraints $c = 1, 1.5, 2, 2.5, 3$, and finally the equal weight portfolio (EQUAL).

For weekly rebalancing, we calculate the annualized portfolio return and annualized out-of-sample standard deviation, given respectively by

$$\widehat{\mu} = 52 \times \frac{1}{40}\sum_{i=11}^{50} \mathbf{w}^{\mathrm{T}}\mathbf{r}_i, \quad \widehat{\sigma} = \Big(52 \times \frac{1}{40}\sum_{i=11}^{50}(\mathbf{w}^{\mathrm{T}}\mathbf{r}_i - \widehat{\mu})^2\Big)^{1/2}.$$

For daily rebalancing case, we just replace the mean weekly return by the mean daily return, and substitute 52 (weeks per year) by 252 (trading days per year). We use similar treatments for the out-of-sample standard deviation. The average of maximum absolute values of portfolio weights over all rebalancing periods and the annual maximum absolute weights are included as well.

Table 1 shows the results for model (5.1) with no factors. It is clear that both NERIVE and EQUAL are similar and are the closest to the theoretical portfolio in terms of the out-of-sample standard deviation, which is a measure of the actual risk. Another notable feature is that NERIVE has the maximum exposure close to the theoretical portfolio as well, at around 1%. This aligns with the result in Theorem 4 that the maximum exposure of NERIVE should be small when p is large. This is in stark contrast with TSCV and other GEC methods, where the smallest average maximum exposure is around 20% for the no-short-sale ($c = 1$) portfolio. For TSCV, it is at 72%, and is a severe over-investment in an individual asset.

Table 2 shows the results for model (5.1) with factors including a pervasive one. The out-of-sample standard deviations of all methods have generally increased compared to the results in Table 1. NERIVE and EQUAL are both better than the theoretical portfolio in terms of this risk measure, and NERIVE and EQUAL perform the best for the 10-day window strategy, while GEC with $c = 2.5$ performs the best for a 10-week training window. NERIVE has annual maximum exposure at 1.1%, which is close to the equal weight portfolio, and is much smaller than the maximum exposure of other methods. For TSCV and all GEC methods, they have invested over 85% in an individual stock in a single period, which is not really an acceptable investment strategy.

Methods	Portfolio Return (%)	Out-of-Sample SD (%)	Aver Max Abs Weight(%)	Ann Max Abs Weight(%)
weekly rebalancing portfolio with 10-week training window				
THEO	0.02	0.9	1.09	1.13
NERIVE	0.17	0.9	1.01	1.02
EQUAL	0.17	0.9	1.00	1.00
TSCV	72.09	18.6	72.02	92.98
GEC1	5.58	3.2	19.79	36.33
GEC1.5	10.60	4.0	23.96	37.11
GEC2	14.28	3.9	30.06	46.13
GEC2.5	13.21	5.9	30.69	47.38
GEC3	18.28	5.3	33.48	47.70
daily rebalancing portfolio with 10-day training window				
THEO	-4.54	2.1	1.20	1.36
NERIVE	0.51	2.1	1.06	1.21
EQUAL	0.23	2.1	1.00	1.00
TSCV	299.75	30.2	46.22	73.97
GEC1	-5.42	7.5	30.39	69.85
GEC1.5	-3.21	10.4	36.90	81.53
GEC2	5.99	12.6	41.87	78.66
GEC2.5	14.32	15.0	44.62	82.21
GEC3	14.89	16.7	51.19	100.31

Table 1: Simulation results for model 1 with no factors ($C = 0$ in (5.1)): Annualized portfolio return, annualized out-of-sample standard deviation, averaged maximum absolute weights and annualized maximum absolute weights for NERIVE, EQUAL, TSCV, and other GEC methods (from $c = 1$ to $c = 3$).

Methods	Portfolio Return (%)	Out-of-Sample SD (%)	Aver Max Abs Weight(%)	Ann Max Abs Weight(%)
weekly rebalancing portfolio with 10-week training window				
THEO	0.5	6.4	5.4	6.6
NERIVE	1.2	6.2	1.0	1.0
EQUAL	1.2	6.2	1.0	1.0
TSCV	108.3	18.4	93.1	140.3
GEC1	7.8	6.5	25.0	40.5
GEC1.5	6.4	5.6	30.3	55.7
GEC2	10.9	5.7	36.3	63.9
GEC2.5	7.6	6.0	34.7	56.2
GEC3	20.4	6.2	40.0	57.8
daily rebalancing portfolio with 10-day training window				
THEO	-5.8	11.7	6.0	8.0
NERIVE	1.9	11.2	1.0	1.1
EQUAL	1.9	11.2	1.0	1.0
TSCV	286.3	37.5	71.4	110.3
GEC1	-7.6	13.2	35.4	85.4
GEC1.5	0.2	14.3	44.7	93.2
GEC2	4.9	16.5	51.4	102.7
GEC2.5	3.3	18.1	58.3	124.0
GEC3	10.4	17.9	62.9	146.5

Table 2: Simulation results for model 2 with factors ($C = 1$ in (5.1)): Annualized portfolio return, annualized out-of-sample standard deviation, averaged maximum absolute weights and annualized maximum absolute weights for NERIVE, EQUAL, TSCV, and other GEC methods (from $c = 1$ to $c = 3$).

We also explicitly measure the difference of portfolio weights of each method and the theoretical weights. To this end, we calculate, for each method,

$$\left\|\widehat{\mathbf{w}} - \mathbf{w}_{\text{theo}}\right\|$$

for each rebalancing period. Figure 1 shows the boxplots of such norms. The case for weekly rebalacing is very similar and hence we do not show the corresponding graphs. The results align with those in Table 1, showing that NERIVE and EQUAL have very close portfolio weights to the theoretical one in each rebalancing period. TSCV performs the worst again, showing that without further regularization, TSCV suffers because of the large bias for the extreme eigenvalues in the estimated integrated covariance matrix.

Table 3 presents the simulation results of three different risk measures for NERIVE, EQUAL, TSCV, and GEC with 5 choices of constraint constants c. Here risk refers to the standard deviation of portfolio returns as in the out-of-sample standard deviation in Tables 1 and 2. We define three risks: theoretical risk $R^{1/2}(\mathbf{w}_{opt})$, actual risk $R^{1/2}(\widehat{\mathbf{w}}_{opt})$, and perceived risk $\widehat{R}^{1/2}(\widehat{\mathbf{w}}_{opt})$, where

$$R^{1/2}(\mathbf{w}_{\text{opt}}) = \sqrt{\mathbf{w}_{\text{opt}}^T \Sigma \mathbf{w}_{\text{opt}}}, \quad R^{1/2}(\widehat{\mathbf{w}}_{opt}) = \sqrt{\widehat{\mathbf{w}}_{\text{opt}}^T \Sigma \widehat{\mathbf{w}}_{\text{opt}}}, \quad \widehat{R}^{1/2}(\widehat{\mathbf{w}}_{\text{opt}}) = \sqrt{\widehat{\mathbf{w}}_{\text{opt}}^T \widehat{\Sigma} \widehat{\mathbf{w}}_{\text{opt}}}.$$

From the table, it is clear that the theoretical risk and the actual risk are very close to each other for NERIVE and EQUAL both with or without factors, which consolidate the results from Table 1 using the

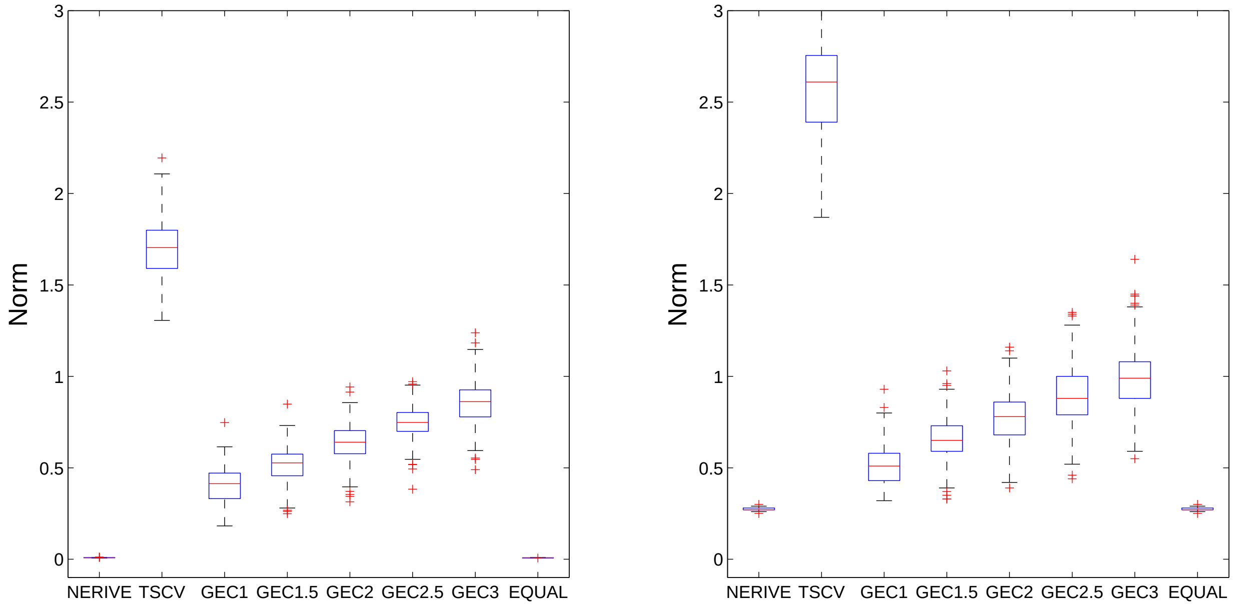

Figure 1: Simulation results: Boxplots of the norms of difference of the theoretical weights and the portfolio weights for NERIVE, TSCV, GEC1, GEC1.5, GEC2, GEC2.5, GEC3, and EQUAL. Investment strategy is daily rebalancing with 10-day training window. Left panel: Without factors ($C = 0$ in (5.1)). Right panel: With factors ($C = 1$ in (5.1)).

out-of-sample standard deviation as the risk measurement. TSCV still gives the highest actual risk, and the no-short-sale portfolio ($c = 1$) performs the best among the GEC methods. A notable feature is that when there are factors, the actual risk for all methods are smaller. In the strategy with a 10-day training window, the actual risks with factors are 2 times smaller for NERIVE and EQUAL, and more than 10 times smaller for other methods.

5.3 Empirical Study

In this study, the choice of the stocks used is based on the "Fifty Most Active Stocks on NYSE, Round Lots (mils. of shares), 2013" and "Fifty Most Active Stocks by Dollar Volume on NYSE ($ in mils.), 2013" from New York Stock Exchange Data official website http://www.nyxdata.com/. There are 26 stocks in both of the two lists above, and 74 stocks in either of the lists. We downloaded all the trading transactions of these 74 stocks in Year 2013 from Wharton Research Data Services (WRDS, https://wrds-web.wharton.upenn.edu/). Due to the lack of price data for Sprint Corporation (Symbol:S), we omit this stock and conduct the empirical study on two settings: 26 stocks and 73 stocks, respectively.

We carry out the same portfolio allocation exercises as in our simulations for both the 26-stock and 73-stock portfolios. Here, we compare two investment strategies. One rebalances weekly using a 10-week

	Without factors ($C = 0$ in (5.1))			With factors ($C = 1$ in (5.1))		
Methods	Theoretical Risk (%)	Actual Risk (%)	Perceived Risk (%)	Theoretical Risk (%)	Actual Risk (%)	Perceived Risk (%)
weekly rebalancing portfolio with 10-week training window						
NERIVE	1.2	1.2	6.9	1.2	1.2	1.0
EQUAL		1.2	6.9		1.2	1.0
TSCV		29.7	19.8		4.2	2.9
GEC1		3.7	42.8		1.2	6.3
GEC1.5		4.8	41.3		1.3	6.1
GEC2		6.0	39.3		1.3	6.0
GEC2.5		6.9	38.2		1.4	5.9
GEC3		7.7	37.3		1.4	5.9
daily rebalancing portfolio with 10-day training window						
NERIVE	1.1	1.1	6.8	.50	.53	.43
EQUAL		1.1	6.8		.53	.43
TSCV		19.0	6.9		1.30	.46
GEC1		4.7	51.7		.57	3.57
GEC1.5		6.0	46.1		.60	3.26
GEC2		7.3	41.1		.63	3.05
GEC2.5		8.4	36.7		.66	2.85
GEC3		9.7	32.7		.69	2.66

Table 3: Simulation results: Theoretical risk, actual risks and perceived risks for NERIVE, EQUAL, TSCV, GEC1, GEC1.5, GEC2, GEC2.5, and GEC3.

training window, another rebalances daily using a 5-day training window. The results are displayed in Tables 4 and 5. In the weekly rebalancing case, we consider the low-frequency method NERCOME as well. The out-of-sample standard deviations are smaller for GEC with small values of constraint c in general. NERIVE and EQUAL are now more different from those in the simulation results, with average maximum exposure of NERIVE being quite different from the equal portfolio weights. The maximum exposure for TSCV and all GEC methods are still much larger than NERIVE and EQUAL, with the no-short-sale portfolio (GEC1) having average maximum exposure of at least 21.2% in either investment strategy and $p = 26$ or $p = 73$. In the 5-day training window strategy for $p = 73$, Table 5 shows that TSCV invested over 760% in an individual stock in an instance, while the no-short-sale portfolio invested 69% in one stock once. These are severe over-investments in individual stocks, while NERIVE, not as low as equal portfolio weights, still has just around 3-4% on average in maximum exposure. The worst sees it invests 7.7% in an individual stock once, which is still a sensible level.

While NERIVE diversifies more than TSCV and GEC methods in both $p = 26$ and $p = 73$ portfolios, it clearly has not overdiversified, as shown by the high level of portfolio returns and Sharpe ratios in general in Tables 4 and 5. Note that the low frequency method NERCOME is actually performing well at a 10-week training window, and even outperforms NERIVE in Sharpe ratio for $p = 73$. The no-short-sale portfolio is also doing well for $p = 23$.

p = 26 Method	Portfolio Return (%)	Out-of-Sample SD(%)	Sharpe Ratio (%)	Aver Max Abs Weight(%)	Ann Max Abs Weight(%)
weekly rebalancing portfolio with 10-week training window					
NERIVE	21.9	10.8	202.4	5.9	8.1
NERCOME	21.3	11.1	192.9	5.1	7.9
EQUAL	20.1	10.5	191.8	3.9	3.9
TSCV	7.2	10.5	68.0	35.1	66.3
GEC1	17.3	8.7	198.3	26.7	41.2
GEC1.5	9.3	9.0	102.6	29.1	44.6
GEC2	7.6	9.7	78.1	32.1	55.0
GEC2.5	6.5	10.2	63.8	34.0	61.0
GEC3	6.8	10.4	65.5	34.9	64.5
daily rebalancing portfolio with 5-day training window					
NERIVE	24.3	11.7	206.8	6.5	14.5
EQUAL	24.2	11.4	213.3	3.9	3.9
TSCV	13.1	12.7	103.1	38.4	77.6
GEC1	25.8	10.5	244.9	28.4	68.7
GEC1.5	18.1	10.2	178.2	31.1	76.7
GEC2	15.1	10.8	139.1	33.7	68.5
GEC2.5	12.8	11.5	111.9	35.7	74.2
GEC3	12.1	12.0	101.3	37.1	76.6

Table 4: Empirical results for the 26 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weights and annualized maximum absolute weights for NERIVE, EQUAL, TSCV, and GEC from constraint $c = 1$ to $c = 3$.

6 Conclusion

We generalize nonlinear shrinkage of eigenvalues in a large sample covariance matrix for independent and identically distributed random vectors (Lam, 2016) to that of a large two-scale covariance matrix estimator (TSCV) for high frequency price data. To do this, we split the data into partitions and regularize the eigenvalues of the TSCV within a partition by the data from other partitions. Regularization is indeed achieved both theoretically and empirically, as demonstrated by the very good performance in our simulations and portfolio allocation exercises.

In using nonlinear shrinkage, one advantage is that we do not need to impose a special structure on the underlying integrated covariance matrix like sparsity or low rank. We get almost sure positive definite integrated covariance matrix estimator asymptotically which is well-conditioned, and this is theoretically as efficient as an ideal estimator while practically performs very well in our simulations and empirical experiments. The induced maximum exposure decay is also clearly demonstrated both theoretically and empirically for our method while other methods show significant over-investment on average to individual stocks. The annualized risk and portfolio weights are close to the theoretical ones in our simulations too.

p = 73 Method	Portfolio Return (%)	Out-of-Sample SD(%)	Sharpe Ratio (%)	Aver Max Abs Weight(%)	Ann Max Abs Weight(%)
		weekly rebalancing portfolio with 10-week training window			
NERIVE	20.7	11.5	180.5	3.2	4.7
NERCOME	24.0	10.4	230.0	2.9	9.1
EQUAL	19.5	11.2	174.6	1.4	1.4
TSCV	1.1	10.6	10.7	26.9	37.9
GEC1	7.8	9.3	83.8	22.4	32.2
GEC1.5	1.9	9.3	20.2	22.4	36.1
GEC2	-4.9	9.4	-52.3	22.9	34.1
GEC2.5	-4.1	10.0	-40.6	23.6	32.3
GEC3	-3.9	10.0	-38.7	24.4	34.5
		daily rebalancing portfolio with 5-day training window			
NERIVE	22.5	12.4	181.3	3.5	7.7
EQUAL	22.3	11.7	189.7	1.4	1.4
TSCV	3.8	13.8	27.7	31.5	760.7
GEC1	15.6	10.2	152.7	21.2	69.0
GEC1.5	16.9	9.6	175.7	23.6	58.0
GEC2	14.9	9.6	154.7	24.3	61.4
GEC2.5	14.8	9.8	150.8	24.4	59.8
GEC3	11.5	10.1	114.3	25.2	55.0

Table 5: Empirical results for the 73 most actively traded stocks in NYSE: annualized portfolio return, annualized out-of-sample standard deviation, Sharpe ratio, averaged maximum absolute weights and annualized maximum absolute weights for NERIVE, EQUAL, TSCV, and GEC from constraint $c = 1$ to $c = 3$.

7 Proof of Theorems

We provide the proof of all the theorems of the paper in this section. Hereafter, we denote $\mathbf{A}\circ\mathbf{B} = (a_{ij}b_{ij})_{i,j}$, the Hadamard product between the matrices $\mathbf{A}$ and $\mathbf{B}$. Define

$$\mathbf{Z}^{\pm}(s) = \mathbf{Y}(s)\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{Y}(s)^{\mathrm{T}}, \quad s = 1, \dots, nL,$$

where $\mathbf{Y}(s)$ is defined in (2.2). We can then write

$$\mathbf{Z}^{\pm}(s) = \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{X}_{v_s}^{\mathrm{T}} + \boldsymbol{\epsilon}_s\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\boldsymbol{\epsilon}_s^{\mathrm{T}}, \tag{7.1}$$

where $\mathbf{X}_{v_s} = (X_{v_s}^{(1)}, \dots, X_{v_s}^{(p)})^{\mathrm{T}}$ is the latent price of all assets at exactly time v_s, and

$$\boldsymbol{\epsilon}_s = \mathbf{Y}(s) - \mathbf{X}_{v_s} = \boldsymbol{\epsilon}(s) + \mathbf{X}(s) - \mathbf{X}_{v_s}.$$

We are now in position to define the multivariate versions of (2.3) and (2.4). For $\ell = 1, \dots, L$, define

$$\begin{aligned} \widehat{\langle \mathbf{Z}^{\pm}, \mathbf{Z}^{\pm} \rangle}_\ell &= [\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)} - \frac{|S^\ell(K)|_K}{|S^\ell(1)|}[\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(1)}, \quad \text{with} \\ [\mathbf{Z}^{\pm}, \mathbf{Z}^{\pm}]_\ell^{(K)} &= \frac{1}{K}\sum_{s\in S^\ell(K)} (\mathbf{Z}^{\pm}(s) - \mathbf{Z}^{\pm}(s-K)) \circ (\mathbf{Z}^{\pm}(s) - \mathbf{Z}^{\pm}(s-K)), \end{aligned} \tag{7.2}$$

where $S^\ell(K)$ and $|S^\ell(K)|_K$ are as defined in (2.3). The TSCV $\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell)$ defined in (2.4) can then be written as

$$\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \frac{1}{4}\left(\langle \widehat{\mathbf{Z}^+, \mathbf{Z}^+}\rangle_\ell - \langle \widehat{\mathbf{Z}^-, \mathbf{Z}^-}\rangle_\ell\right). \tag{7.3}$$

Define

$$\mathbf{X}_s^\pm = \mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\mathbf{X}_{v_s}^{\mathrm{T}}, \quad \mathbf{E}_s^\pm = \boldsymbol{\epsilon}_s\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\boldsymbol{\epsilon}_s^{\mathrm{T}},$$

so that $\mathbf{Z}^\pm(s)$ in (7.1) can be written as $\mathbf{Z}^\pm(s) = \mathbf{X}_s^\pm + \mathbf{E}_s^\pm$. Then for each positive integer K, we can write $[\mathbf{Z}^\pm, \mathbf{Z}^\pm]_\ell^{(K)}$ in (7.2) as

$$[\mathbf{Z}^\pm, \mathbf{Z}^\pm]_\ell^{(K)} = [\mathbf{X}^\pm, \mathbf{X}^\pm]_\ell^{(K)} + 2[\mathbf{X}^\pm, \mathbf{E}^\pm]_\ell^{(K)} + [\mathbf{E}^\pm, \mathbf{E}^\pm]_\ell^{(K)}, \tag{7.4}$$

where

$$[\mathbf{X}^\pm, \mathbf{E}^\pm]_\ell^{(K)} = \frac{1}{K}\sum_{s\in S^\ell(K)} (\mathbf{X}_s^\pm - \mathbf{X}_{s-K}^\pm) \circ (\mathbf{E}_s^\pm - \mathbf{E}_{s-K}^\pm),$$

and $[\mathbf{X}^\pm, \mathbf{X}^\pm]_\ell^{(K)}$ and $[\mathbf{E}^\pm, \mathbf{E}^\pm]_\ell^{(K)}$ are defined similarly. Hence we can decompose $\widetilde{\mathbf{\Sigma}}(\tau_{\ell-1}, \tau_\ell) = \sum_{i=1}^6 \mathbf{B}_i^\ell$, where

$$\begin{aligned}
\mathbf{B}_1^\ell &= \frac{1}{4}\left([\mathbf{X}^+, \mathbf{X}^+]_\ell^{(K)} - [\mathbf{X}^-, \mathbf{X}^-]_\ell^{(K)}\right), \\
\mathbf{B}_2^\ell &= -\frac{|S^\ell(K)|_K}{4|S^\ell(1)|}\left([\mathbf{X}^+, \mathbf{X}^+]_\ell^{(1)} - [\mathbf{X}^-, \mathbf{X}^-]_\ell^{(1)}\right), \\
\mathbf{B}_3^\ell &= \frac{1}{2}\left([\mathbf{X}^+, \mathbf{E}^+]_\ell^{(K)} - [\mathbf{X}^-, \mathbf{E}^-]_\ell^{(K)}\right), \\
\mathbf{B}_4^\ell &= -\frac{|S^\ell(K)|_K}{2|S^\ell(1)|}\left([\mathbf{X}^+, \mathbf{E}^+]_\ell^{(1)} - [\mathbf{X}^-, \mathbf{E}^-]_\ell^{(1)}\right), \\
\mathbf{B}_5^\ell &= \frac{1}{4}\left([\mathbf{E}^+, \mathbf{E}^+]_\ell^{(K)} - [\mathbf{E}^-, \mathbf{E}^-]_\ell^{(K)}\right), \\
\mathbf{B}_6^\ell &= -\frac{|S^\ell(K)|_K}{4|S^\ell(1)|}\left([\mathbf{E}^+, \mathbf{E}^+]_\ell^{(1)} - [\mathbf{E}^-, \mathbf{E}^-]_\ell^{(1)}\right).
\end{aligned} \tag{7.5}$$

We now present a series of Lemmas before proving Theorem 1.

Lemma 1. *Let Assumptions (A1) to (A6) hold. Then for unit vectors $\mathbf{x}_{ij}$ being independent of $\{\mathbf{X}_{v_s}\}_{s\in S^j(1)}$ for $i = 1, \ldots, p$ and $j = 1, \ldots, L$, we have for $\mathbf{B}_1^j$ defined in (7.5),*

$$\max_{\substack{i=1,\ldots,p \\ 1\le j\le L}} \left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}} - 1\right| \xrightarrow{a.s.} 0.$$

At the same time, we have

$$\max_{\substack{i=1,\ldots,p \\ 1\le j\le L}} \left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_2^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}}\right| \xrightarrow{a.s.} 0.$$

Proof of Lemma 1. Define $\mathbf{D}_x$ as the diagonal matrix with the elements in a vector $\mathbf{x}$ as the diagonal, and that $\Delta_K \mathbf{X}_{v_s} = \mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}}$. Using the identity $\mathbf{x}^{\mathrm{T}}\mathbf{A}\circ\mathbf{B}\mathbf{y} = \mathrm{tr}(\mathbf{D}_x\mathbf{A}\mathbf{D}_y\mathbf{B}^{\mathrm{T}})$, with $\mathrm{tr}(\cdot)$ being the trace of a matrix, we have for any $\mathbf{x}\in\mathbb{R}^p$ and K a positive integer,

$$\begin{aligned}
\mathbf{x}^{\mathrm{T}}[\mathbf{X}^{\pm},\mathbf{X}^{\pm}]_j^{(K)}\mathbf{x} &= \frac{1}{K}\sum_{s\in S^j(K)} \mathbf{x}^{\mathrm{T}}(\mathbf{X}_s^{\pm}-\mathbf{X}_{s-K}^{\pm})\circ(\mathbf{X}_s^{\pm}-\mathbf{X}_{s-K}^{\pm})\mathbf{x} \\
&= \pm\frac{1}{K}\sum_{s\in S^j(K)} \mathrm{tr}\big(\mathbf{D}_x(\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}})\big)^2 \\
&= \pm\frac{1}{K}\sum_{s\in S^j(K)} \mathrm{tr}\Big(\mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}}\mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} \pm \mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}}\mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}} \\
&\qquad \pm \mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\mathbf{D}_x\Delta_K\mathbf{X}_{v_s}\mathbf{1}_p^{\mathrm{T}} + \mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\mathbf{D}_x\mathbf{1}_p\Delta_K\mathbf{X}_{v_s}^{\mathrm{T}}\Big) \\
&= \pm\frac{2}{K}\sum_{s\in S^j(K)} \Big((\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_s})^2 \pm \sum_i x_i\sum_i x_i(\Delta_K\mathbf{X}_{v_s})_i^2\Big).
\end{aligned} \tag{7.6}$$

This implies that

$$\mathbf{x}^{\mathrm{T}}\mathbf{B}_1\mathbf{x} = \frac{1}{K}\sum_{s\in S^j(K)}(\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_s})^2 = \frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K}(\mathbf{x}^{\mathrm{T}}\Delta_K\mathbf{X}_{v_{rK+\ell}})^2. \tag{7.7}$$

In the above, $v_{rK+\ell} = v^j_{rK+\ell}$, where $\tau_{j-1} = v_0^j < v_1^j < \cdots < v^j_{n(j)} \leq \tau_j$ are all the all-refresh time points in the partition $(\tau_{j-1},\tau_j]$. Hereafter, the superscript j will be suppressed if there are no ambiguity arise for simpler notations. By the diffusion model (2.1), we can write

$$\Delta_K\mathbf{X}_{v_{rK+\ell}} = \mathbf{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})^{1/2}\mathbf{Z}^j_{r,\ell}, \tag{7.8}$$

where the vectors $\mathbf{Z}^j_{r,\ell}$'s are independent of each other for different values of $r = 1,\ldots,|S^j(K)|_K$ and $j=1,\ldots,L$. Each $\mathbf{Z}^j_{r,\ell}$ has independent entries each with mean 0 and variance 1. Hence, combining (7.7) and (7.8), we have

$$\max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_1^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}} - 1\right| \leq \max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K} g^j_{r,\ell}\right| + R_1,$$

where $g^j_{r,\ell}$ and R_1 are defined as

$$g^j_{r,\ell} = \frac{(\mathbf{Z}^{j\mathrm{T}}_{r,\ell}\mathbf{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})^{1/2}\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}, \tag{7.9}$$

$$R_1 = \max_{\substack{i=1,\ldots,p\\1\leq j\leq L}}\left|\frac{\frac{1}{K}\sum_{\ell=0}^{K-1}\Big(\sum_{r=1}^{|S^j(K)|_K}\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell})\mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}\Big)}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|. \tag{7.10}$$

To bound R_1, using Assumptions (A2) and (A4),

$$
\begin{aligned}
R_1 &\le \frac{L}{C_1C_3K} \max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \Big| \sum_{\ell=0}^{K-1} \Big(\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(v_\ell, v_{|S^j(K)|-K+1+\ell}) \mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{x}_{ij} \Big) \Big| \\
&= \frac{L}{C_1C_3K} \max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \sum_{\ell=0}^{K-1} \Big(\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(\tau_{j-1}, v_\ell) \mathbf{x}_{ij} + \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(v_{|S^j(K)|-K+1+\ell}, \tau_j) \mathbf{x}_{ij} \Big) \\
&\le \frac{L}{C_1C_3K} \sum_{\ell=0}^{K-1} \Big(\frac{C_2C_5\ell}{nL} + \frac{C_2C_5(n(j)-|S^j(K)|+K-1-\ell)}{nL} \Big) + \frac{C_2}{C_1C_3} o(1) \\
&= O(K/n) + o(1) = o(1),
\end{aligned}
$$

since $|S^j(K)| = n(j) - K$. To bound the first term, we first note that $E(g_{r,\ell}^j|\mathbf{x}) = 0$. Using Lemma 2.7 of Bai and Silverstein (1998), for any integer $q \ge 2$,

$$
\begin{aligned}
E(|g_{r,\ell}^j|^q|\mathbf{x}) &\le \frac{L^qK_q}{C_1^qC_3^q} \Big((E^{q/2}|z_1|^4 + E|z_1|^{2q})(\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(v_{(r-1)K+\ell}, v_{rK+\ell}) \mathbf{x}_{ij})^q \Big) \\
&= O(K^q n^{-q}) = O(n^{-q/3}),
\end{aligned}
\tag{7.11}
$$

where we used the fact the all moments are finite for the components in $\mathbf{Z}_{r,\ell}^j$ since they are normally distributed by the assumption of the diffusion model (2.1), and we used Assumptions (A4) and (A6) in the last line. Then we have for $q \ge 1$,

$$
\begin{aligned}
E\Bigg(\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \Big(\frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} g_{r,\ell}^j \Big)^{2q} |\mathbf{x} \Bigg) &\le E\Bigg(\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \max_{1\le \ell\le K-1} \Big(\sum_{r=1}^{|S^j(K)|_K} g_{r,\ell}^j \Big)^{2q} |\mathbf{x} \Bigg) \\
&= O\Big(pLK \sum_{r_1\ne\cdots\ne r_q} \prod_{k=1}^{q} E((g_{r_k,\ell}^j)^2|\mathbf{x}) \Big) \\
&= O(pLK \cdot |S^j(K)|_K^q \cdot n^{-2q/3}) = O(n^{-q/3+68/39}),
\end{aligned}
$$

where we used the fact that the largest order from the expansion in the first line must comes from the product of all square terms. Hence we can set q with $q \ge 9$, so that $\sum_{n\ge 1} n^{-q/3+68/39} < \infty$, proving that $\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \Big| \frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} g_{r,\ell}^j \Big| \xrightarrow{a.s.} 0$ through the Borel-Cantelli lemma. This proves the first part of Lemma 1.

For the second part, applying (7.6) and (7.7) with $K = 1$, we have

$$
\max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \Bigg| \frac{\mathbf{x}_{ij}^{\mathrm{T}} \mathbf{B}_2^j \mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}(\tau_{j-1}, \tau_j) \mathbf{x}_{ij}} \Bigg| \le \max_{\substack{i=1,\ldots,p\\ 1\le j\le L}} \Bigg| \frac{|S^j(K)|_K}{|S^j(1)|} \sum_{r=1}^{S^j(1)} g_{r,0}^j \Bigg| + R_2,
$$

where $g^j_{r,0}$ is defined as in (7.9) with $K = 1$, and R_2 is defined as and bounded by

$$R_2 = \max_{\substack{i=1,\dots,p\\1\le j\le L}} \frac{L|S^j(K)|_K}{C_1C_3|S^j(1)|}\left|\sum_{r=1}^{|S^j(1)|} \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(v_{r-1}, v_r)\mathbf{x}_{ij}\right| \le \max_{\substack{i=1,\dots,p\\1\le j\le L}} \frac{L|S^j(K)|_K}{C_1C_3|S^j(1)|}\frac{C_2C_4}{L} = O(K^{-1}) = o(1).$$

Finally, $E(|g^j_{r,0}|^q|\mathbf{x}) = O(n^{-q})$ for $q \ge 2$ by (7.11), and

$$E\Big(\max_{\substack{i=1,\dots,p\\1\le j\le L}} \Big(\frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{|S^j(1)|} g^j_{r,0}\Big)^{2q}\Big) = O(pLK^{-2q}|S^j(1)|^q n^{-2q}) = O(n^{-7q/3+14/13}),$$

so that $q = 1$ here is sufficient for showing $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{|S^j(K)|_K}{|S^j(1)|}\sum_{r=1}^{S^j(1)} g^j_{r,0}\right| \xrightarrow{a.s.} 0$. This completes the proof of the lemma. $\square$

Lemma 2. *Let Assumptions (A1) to (A6) hold. Then for unit vectors $\mathbf{x}_{ij}$ being independent of $\{\mathbf{X}_{v_s}\}_{s\in S^j(1)}$ and $\{\boldsymbol{\epsilon}_s\}_{s\in S^j(1)}$ for $i = 1,\dots,p$ and $j = 1,\dots,L$, we have for $\mathbf{B}^j_3, \mathbf{B}^j_4$ defined in (7.5),*

$$\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}^j_3 + \mathbf{B}^j_4)\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{x}_{ij}}\right| \xrightarrow{a.s.} 0.$$

Proof of Lemma 2. Similar to the arguments in (7.6), we can show that

$$\frac{1}{2}(\mathbf{x}_{ij}^{\mathrm{T}}[\mathbf{Z}^+, \mathbf{E}^+]^{(K)}_j\mathbf{x}_{ij} - \mathbf{x}_{ij}^{\mathrm{T}}[\mathbf{Z}^-, \mathbf{E}^-]^{(K)}_j\mathbf{x}_{ij}) = \frac{2}{K}\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\boldsymbol{\epsilon}_s - \boldsymbol{\epsilon}_{s-K})$$
$$= \frac{2}{K}\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\boldsymbol{\epsilon}(s) - \boldsymbol{\epsilon}(s-K) - (\mathbf{X}_{v_s} - \mathbf{X}(s)) + (\mathbf{X}_{v_{s-K}} - \mathbf{X}(s-K))).$$

Hence we can decompose $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}^j_3\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right| \le R_\epsilon - R^{(K)}_\epsilon - R_x + R^{(K)}_x$, where

$$R_\epsilon = \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s)\right|,$$
$$R^{(K)}_\epsilon = \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s-K)\right|,$$
$$R_x = \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_s} - \mathbf{X}(s))\right|,$$
$$R^{(K)}_x = \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{s\in S^j(K)}(\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_{s-K}} - \mathbf{X}(s-K))\right|.$$

Define, for $s \in S^j(K)$, $g_{j,s} = (\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s)$. Then since $\boldsymbol{\epsilon}(\cdot)$ is independent of $\mathbf{X}(\cdot)$, we have

$E(g_{j,s}|\mathbf{x}) = 0$ and the $g_{j,s}|\mathbf{x}$'s are uncorrelated with each other for fixed j. We also have for $q \geq 2$,

$$\begin{aligned}
E(|g_{j,s}|^q|\mathbf{x}) &\leq E^{1/2}\big((\mathbf{X}_{v_s} - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}|\mathbf{x}\big)^{2q} E^{1/2}\big(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}(s)\big|\mathbf{x}\big)^{2q} \\
&= E^{1/2}\Big(\mathbf{Z}_s^{j\mathrm{T}}\boldsymbol{\Sigma}(v_{s-K}, v_s)^{1/2}\mathbf{x}_{ij}|\mathbf{x}\Big)^{2q} E^{1/2}\big(\mathbf{Z}_{\epsilon,s}^{j\mathrm{T}}\boldsymbol{\Sigma}_\epsilon^{1/2}\mathbf{x}_{ij}|\mathbf{x}\big)^{2q} \\
&= O\big((\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(v_{s-K}, v_s)\mathbf{x}_{ij})^{q/2}(\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}_\epsilon\mathbf{x}_{ij})^{q/2}\big) = O(K^{q/2}n^{-q/2}L^{-q/2}),
\end{aligned}$$

where the $\mathbf{Z}_s^j$ and $\mathbf{Z}_{\epsilon,s}^j$ are independent of each other for each s, with independent normal entries by the diffusion model (2.1) and Assumption (A3). With this, for integer $q \geq 1$,

$$\begin{aligned}
E|R_\epsilon|^{2q} &= O\bigg(L^{2q}K^{-2q} \cdot pL \cdot \Big(\sum_{s \in S^j(K)} g_{j,s}\Big)^{2q}\bigg) = O(pL^{2q+1}K^{-2q}|S^j(K)|^q \cdot K^q n^{-q}L^{-q}) \\
&= O(pL^{q+1}K^{-q}) = O(n^{-23q/39+14/13}),
\end{aligned}$$

and we can set $q = 3$ so that $\sum_{n\geq 1} n^{-23q/39+14/13} < \infty$, proving via the Borel-Cantelli lemma that $R_\epsilon \xrightarrow{a.s.} 0$. Similar arguments show that $R_\epsilon^{(K)} \xrightarrow{a.s.} 0$ also.

Consider $R_x \leq R_{x,1} + R_{x,2}$, where

$$\begin{aligned}
R_{x,1} &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big|\sum_{s\in S^j(K)} ((\mathbf{X}_{v_s} - \mathbf{X}(s))^{\mathrm{T}}\mathbf{x}_{ij})^2\Big|, \\
R_{x,2} &= \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big|\sum_{s\in S^j(K)} (\mathbf{X}(s) - \mathbf{X}_{v_{s-K}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v_s} - \mathbf{X}(s))\Big|.
\end{aligned}$$

To analyze both terms, consider the previous-tick time $t_s^i \in (v_{s-1}, v_s]$ for the ith asset, which should satisfies

$$v_{s-1} < t_s^{(i_1)} \leq t_s^{(i_2)} \leq \cdots \leq t_s^{(i_p)} = v_s,$$

where $i_1, \ldots, i_p$ is some permutation of $1, \ldots, p$. Letting b_s denotes the number of tides, we can write the above as

$$v_{s-1} < t_s^{j_1} < t_s^{j_2} < \cdots < t_s^{j_{p-b_s}} = v_s,$$

where $j_1, \ldots, j_{p-b_s} \in \{1, \ldots, p\}$.

Then we can write, for $s = 1, \ldots, nL$,

$$\mathbf{X}_{v_s} - \mathbf{X}(s) = \sum_{k=1}^{p-b_s-1} \mathbf{D}_k^s\boldsymbol{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})^{1/2}\mathbf{Z}_k^s, \tag{7.12}$$

where $\mathbf{D}_k^s$ is a diagonal matrix with either 0 or 1 as elements. The jth diagonal element is 1 if the jth asset is already traded at time $t_s^{j_k}$, and 0 otherwise. The $\mathbf{Z}_k^s$'s are independent of each other for $k = 1, \ldots, p-b_s-1$, each with independent standard normal entries. Define

$$g_{ij}^s = ((\mathbf{X}_{v_s} - \mathbf{X}(s))^{\mathrm{T}}\mathbf{x}_{ij})^2 - \sum_{k=1}^{p-b_s-1} \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_k^s\mathbf{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})\mathbf{D}_k^s\mathbf{x}_{ij} = \sum_{k_1,k_2}^{p-b_s-1} h_{k_1k_2,ij}^s, \quad \text{where}$$

$$h_{k_1k_2,ij}^s = \begin{cases} \mathbf{Z}_{k_1}^{s\mathrm{T}}\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})^{1/2}\mathbf{D}_{k_1}^s\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_2}^s\mathbf{\Sigma}(t_s^{j_{k_2}}, t_s^{j_{k_2+1}})^{1/2}\mathbf{Z}_{k_2}^s, & k_1 \neq k_2; \\ (\mathbf{Z}_{k_1}^{s\mathrm{T}}\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})^{1/2}\mathbf{D}_{k_1}^s\mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_1}^s\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})\mathbf{D}_{k_1}^s\mathbf{x}_{ij}, & k_1 = k_2. \end{cases}$$

which have $E(g_{ij}^s|\mathbf{x}) = E(h_{k_1k_2,ij}^s|\mathbf{x}) = 0$, and each g_{ij}^s is uncorrelated of each other for $s \in S^j(K)$, while $h_{k_1k_2,ij}^s$ is uncorrelated of each other as long as either the index k_1 or k_2 are different, for fixed i, j and s. Then for $q \geq 2$ and $k_1 \neq k_2$,

$$E(|h_{k_1k_1,ij}^s|^q|\mathbf{x}) = O((\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_1}^s\mathbf{\Sigma}(t_s^{j_{k_1}}, t_s^{j_{k_1+1}})\mathbf{D}_{k_1}^s\mathbf{x}_{ij})^q) = O(n^{-q}L^{-q}(p-b_s-1)^{-q}),$$
$$E(|h_{k_1k_2,ij}^s|^q|\mathbf{x}) \leq \prod_{r=1}^{2} E^{1/2}((h_{k_rk_r,ij}^s + (\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_{k_r}^s\mathbf{\Sigma}(t_s^{j_{k_r}}, t_s^{j_{k_r+1}})\mathbf{D}_{k_r}^s\mathbf{x}_{ij}))^q|\mathbf{x}) = O(n^{-q}L^{-q}(p-b_s-1)^{-q}),$$

where the last line used the result in the first line. Then we have for $q \geq 2$,

$$E(|g_{ij}^s|^q|\mathbf{x}) \leq E^{1/2}(|g_{ij}^s|^{2q}|\mathbf{x}) = O(((p-b_s-1)^{2q} \cdot n^{-2q}L^{-2q}(p-b_s-1)^{-2q})^{1/2}) = O(n^{-q}L^{-q}). \quad (7.13)$$

With this result, we can now decompose

$$R_{x,1} = \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \sum_{s\in S^j(K)} g_{ij}^s \right| + R, \quad \text{where}$$
$$R = \frac{2L}{C_1C_3K} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \sum_{s\in S^j(K)} \sum_{k=1}^{p-b_s-1} \mathbf{x}_{ij}^{\mathrm{T}}\mathbf{D}_k^s\mathbf{\Sigma}(t_s^{j_k}, t_s^{j_{k+1}})\mathbf{D}_k^s\mathbf{x}_{ij} \right| = O(1/K) = o(1).$$

The first term has

$$E((R_{x,1} - R)^{2q}|\mathbf{x}) = O(pL \cdot L^{2q}K^{-2q}|S^j(K)|^q \cdot n^{-2q}L^{-2q}) = O(n^{-q+1}LK^{-2q}) = O(n^{-7q/3+14/13}),$$

so that we can set $q = 1$ and through the Borel-Cantelli lemma, $R_{x,1} - R \overset{a.s.}{\to} 0$, and so $R_{x,1} \overset{a.s.}{\to} 0$ since $R = o(1)$.

For the term $R_{x,2}$, it can be bounded as

$$R_{x,2} = \frac{2L}{C_1C_3K}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^j(K)|_K}(\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell))\right|$$
$$\le \frac{2L}{C_1C_3}\max_{\substack{i=1,\dots,p\\1\le j\le L}}\max_{0\le\ell\le K-1}\left|\sum_{r=1}^{|S^j(K)|_K}g^j_{r,\ell}\right|, \text{ where}$$
$$g^j_{r,\ell} = (\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)),$$

with $\mathbf{X}^j(s)$ representing the process $\mathbf{X}$ in the jth partition, and the index s goes from 0 to $|S^j(K)|$. Note that we have $E(g^j_{r,\ell}|\mathbf{x})=0$, and each $g^j_{r,\ell}$ is independent of each other for fixed j,ℓ and $r=1,\dots,|S^j(K)|_K$. For $q\ge 2$, we then have

$$E(|g^j_{r,\ell}|^q|\mathbf{x}) \le E^{1/2}(((\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x})E^{1/2}((\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)))^{2q}|\mathbf{x})$$
$$\le 2^{(q-1)/2}\Big(\mathbf{E}^{1/2}\big(((\mathbf{X}^j(rK+\ell)-\mathbf{X}_{v^j_{rK+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x}\big)+E^{1/2}\big(((\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}_{v^j_{(r-1)K+\ell}})^{\mathrm{T}}\mathbf{x}_{ij})^{2q}|\mathbf{x}\big)\Big)$$
$$\cdot E^{1/2}((\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{X}_{v^j_{rK+\ell}}-\mathbf{X}^j(rK+\ell)))^{2q}|\mathbf{x})$$
$$= O((n^{-q}L^{-q})^{1/2}+(K^qn^{-q}L^{-q})^{1/2})\cdot O((n^{-q}L^{-q})^{1/2}) = O(K^{q/2}n^{-q}L^{-q}),$$

where we used the results from (7.11) and (7.13) in the last line. Then for $q\ge 1$,

$$E(R^{2q}_{x,2}|\mathbf{x}) = O(pLK\cdot L^{2q}|S^j(K)|^q_K K^q n^{-2q}L^{-2q}) = O(LKn^{-q+1}) = O(n^{-q+68/39}),$$

so that we can set $q=2$, and through the Borel-Cantelli lemma to show that $R_{x,2}\xrightarrow{a.s.}0$. Hence we finally have $R_x\le R_{x,1}+R_{x,2}\xrightarrow{a.s.}0$. Similarly, we can show using the same set of arguments that $R_x^{(K)}\xrightarrow{a.s.}0$, so that $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_3^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$. Note that the same arguments used in the proof of this lemma can be applied to show that $\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}\mathbf{B}_4^j\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0$. This completes the proof of the lemma. □

Lemma 3. *Let Assumptions (A1) to (A6) hold. Then for unit vectors $\mathbf{x}_{ij}$ being independent of $\{\boldsymbol{\epsilon}_s\}_{s\in S^j(1)}$ for $i=1,\dots,p$ and $j=1,\dots,L$, we have for $\mathbf{B}_5^j,\mathbf{B}_6^j$ defined in (7.5),*

$$\max_{\substack{i=1,\dots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}_5^j+\mathbf{B}_6^j)\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\mathbf{x}_{ij}}\right|\xrightarrow{a.s.}0.$$

Proof of Lemma 3. Similar to the arguments in (7.6), we can show that

$$\frac{1}{4}\mathbf{x}_{ij}^{\mathrm{T}}([\mathbf{E}^{+},\mathbf{E}^{+}]_{j}^{(K)}-[\mathbf{E}^{-},\mathbf{E}^{-}]_{j}^{(K)})\mathbf{x}_{ij}=\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^{j}(K)|_{K}}\big((\boldsymbol{\epsilon}_{rK+\ell}^{j}-\boldsymbol{\epsilon}_{(r-1)K+\ell}^{j})^{\mathrm{T}}\mathbf{x}_{ij}\big)^{2}$$

$$=\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^{j}(K)|_{K}}\big((\boldsymbol{\epsilon}^{j}(rK+\ell)-\boldsymbol{\epsilon}^{j}((r-1)K+\ell)$$

$$+(\mathbf{X}^{j}(rK+\ell)-\mathbf{X}_{v_{rK+\ell}^{j}})+(\mathbf{X}_{v_{(r-1)K+\ell}^{j}}-\mathbf{X}^{j}((r-1)K+\ell)))^{\mathrm{T}}\mathbf{x}_{ij}\big)^{2},$$

where $\boldsymbol{\epsilon}^{j}(s)$ is the process $\boldsymbol{\epsilon}(\cdot)$ in the jth partition, with $s=0,\ldots,|S^{j}(K)|$. This implies that we have

$$\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{\mathbf{x}_{ij}^{\mathrm{T}}(\mathbf{B}_{5}^{j}+\mathbf{B}_{6}^{j})\mathbf{x}_{ij}}{\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\Sigma}(\tau_{j-1},\tau_{j})\mathbf{x}_{ij}}\right|\le I_{1}+I_{2}+I_{3}+R,\quad\text{where}$$

$$I_{1}=\frac{L}{C_{1}C_{3}}\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^{j}(K)|_{K}}(\boldsymbol{\epsilon}^{j}(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij})^{2}-\frac{|S^{j}(K)|_{K}}{|S^{j}(1)|}\sum_{r=1}^{|S^{j}(1)|}(\boldsymbol{\epsilon}^{j}(r)^{\mathrm{T}}\mathbf{x}_{ij})^{2}\right|,$$

$$I_{2}=\frac{L}{C_{1}C_{3}}\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^{j}(K)|_{K}}(\boldsymbol{\epsilon}^{j}((r-1)K+\ell)^{\mathrm{T}}\mathbf{x}_{ij})^{2}-\frac{|S^{j}(K)|_{K}}{|S^{j}(1)|}\sum_{r=1}^{|S^{j}(1)|}(\boldsymbol{\epsilon}^{j}(r-1)^{\mathrm{T}}\mathbf{x}_{ij})^{2}\right|,$$

$$I_{3}=\frac{2L}{C_{1}C_{3}}\max_{\substack{i=1,\ldots,p\\1\le j\le L}}\left|\frac{1}{K}\sum_{\ell=0}^{K-1}\sum_{r=1}^{|S^{j}(K)|_{K}}\boldsymbol{\epsilon}^{j}(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}^{j}((r-1)K+\ell)\right.$$

$$\left.-\frac{|S^{j}(K)|_{K}}{|S^{j}(1)|}\sum_{r=1}^{|S^{j}(1)|}\boldsymbol{\epsilon}^{j}(r)^{\mathrm{T}}\mathbf{x}_{ij}\mathbf{x}_{ij}^{\mathrm{T}}\boldsymbol{\epsilon}^{j}(r-1)\right|,$$

and R is the absolute sum of terms involving all other products, including those between $\boldsymbol{\epsilon}^{j}(rK+\ell)^{\mathrm{T}}\mathbf{x}_{ij}$ or $\boldsymbol{\epsilon}^{j}((r-1)K+\ell)^{\mathrm{T}}\mathbf{x}_{ij}$ and $(\mathbf{X}^{j}(rK+\ell)-\mathbf{X}_{v_{rK+\ell}^{j}})^{\mathrm{T}}\mathbf{x}_{ij}$ or $(\mathbf{X}^{j}((r-1)K+\ell)-\mathbf{X}_{v_{(r-1)K+\ell}^{j}})^{\mathrm{T}}\mathbf{x}_{ij}$ (terms contributed from $\mathbf{B}_{6}^{j}$ has $K=1$ and $\ell=0$). These terms can be proved to be asymptotically 0 almost surely with techniques similar to those employed in proving R_{ϵ}, $R_{\epsilon}^{(K)}$ and $R_{x}\overset{a.s.}{\to}0$ in the proof of Lemma 2. For the products between $(\mathbf{X}^{j}(rK+\ell)-\mathbf{X}_{v_{rK+\ell}^{j}})^{\mathrm{T}}\mathbf{x}_{ij}$ and $(\mathbf{X}^{j}((r-1)K+\ell)-\mathbf{X}_{v_{(r-1)K+\ell}^{j}})^{\mathrm{T}}\mathbf{x}_{ij}$, they can also be proved to be asymptotically 0 almost surely using techniques similar to proving R_{x} and $R_{x}^{(K)}\overset{a.s.}{\to}0$ in the proof of Lemma 2. Hence we have $R\overset{a.s.}{\to}0$ and we omit the details.

We can write $\boldsymbol{\epsilon}^{j}(rK+\ell)=\boldsymbol{\Sigma}_{\epsilon}^{1/2}\mathbf{Z}_{r,\ell}^{j}$, where $\mathbf{Z}_{r,\ell}^{j}$ is independent of each other for $r=1,\ldots,|S^{j}(K)|_{K}$, $\ell=0,\ldots,K-1$ and fixed K, with independent standard normal entries. Then we can decompose $I_{1}\le$

$I_{1,1} + I_{1,2}$, where

$$I_{1,1} = \frac{L}{C_1C_3} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big| \frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} g_{r,\ell}^j \Big|, \;\; \text{with } g_{r,\ell}^j = (\mathbf{Z}_{r,\ell}^{j\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon \mathbf{x}_{ij},$$

$$I_{1,2} = \frac{L}{C_1C_3} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \frac{|S^j(K)|_K}{|S^j(1)|} \Big| \sum_{r=1}^{|S^j(1)|} g_{r,0}^j \Big|, \;\; \text{with } g_{r,0}^j = (\mathbf{Z}_{r,0}^{j\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{x}_{ij})^2 - \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon \mathbf{x}_{ij}.$$

Each $g_{r,\ell}^j$ is independent of each other for different r, ℓ and fixed j and K, with $E(g_{r,\ell}^j|\mathbf{x}) = 0$ and $E(|g_{r,\ell}^j|^q|\mathbf{x}) = O((\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon \mathbf{x}_{ij})^q) = O(1)$. Hence for $q \geq 1$,

$$E(I_{1,1}^{2q}|\mathbf{x}) = O(pL \cdot L^{2q} K^{-2q} \cdot (|S^j(K)| - K + 1)^q) = O(n^{-7q/39+14/13}),$$

so that with $q = 12$, we have $I_{1,1} \stackrel{a.s.}{\to} 0$ through the Borel-Cantelli lemma. Exactly the same arguments show $I_{1,2} \stackrel{a.s.}{\to} 0$. Hence $I_1 \stackrel{a.s.}{\to} 0$. We also have $I_2 \stackrel{a.s.}{\to} 0$ by the same decomposition and arguments for I_1.

Finally, we can decompose $I_3 \leq I_{3,1} + I_{3,2}$, where

$$I_{3,1} = \frac{2L}{C_1C_3} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big| \frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{r=1}^{|S^j(K)|_K} \mathbf{Z}_{r,\ell}^{j\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{x}_{ij} \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{Z}_{r-1,\ell}^j \Big|,$$

$$I_{3,2} = \frac{2L}{C_1C_3} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big| \frac{|S^j(K)|_K}{|S^j(1)|} \sum_{r=1}^{|S^j(1)|} \mathbf{Z}_{r,0}^{j\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{x}_{ij} \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{Z}_{r-1,\ell}^j \Big|.$$

To find the order of $I_{3,1}$, we can decompose $I_{3,1} \leq I_{3,1,\mathrm{even}} + I_{3,1,\mathrm{odd}}$, so for instance,

$$I_{3,1,\mathrm{even}} = \frac{2L}{C_1C_3} \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \Big| \frac{1}{K} \sum_{\ell=0}^{K-1} \sum_{\substack{r=1 \\ r \text{ even}}}^{|S^j(K)|_K} g_{r,\ell}^j \Big|, \;\; \text{where } g_{r,\ell}^j = \mathbf{Z}_{r,\ell}^{j\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{x}_{ij} \mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{Z}_{r-1,\ell}^j.$$

Then each $g_{r,\ell}^j$ is independent of each other for r even and for each ℓ and fixed j, with $E(g_{r,\ell}^j|\mathbf{x}) = 0$ and for $q \geq 2$, $E(|g_{r,\ell}^j|^q|\mathbf{x}) = O((\mathbf{x}_{ij}^{\mathrm{T}} \boldsymbol{\Sigma}_\epsilon \mathbf{x}_{ij})^q) = O(1)$. Hence

$$E(|I_{3,1,\mathrm{even}}|^{2q}|\mathbf{x}) = O(pL \cdot L^{2q} K^{-2q} [|S^j(K)|_K/2 \cdot K]^q) = O(n^{-7q/39+14/13}),$$

so that at $q = 12$ we have $I_{3,1,\mathrm{even}} \stackrel{a.s.}{\to} 0$. Same arguments show that $I_{3,1,\mathrm{odd}} \stackrel{a.s.}{\to} 0$, and hence $I_{3,1} \stackrel{a.s.}{\to} 0$. We can decompose $I_{3,2}$ in the same way to show that $I_{3,2} \stackrel{a.s.}{\to} 0$. Therefore, we have $I_3 \stackrel{a.s.}{\to} 0$. This completes the proof of the lemma. □

Proof of Theorem 1. By the independent increment property of the diffusion process in (2.1) for $\{\mathbf{X}_t\}$, and the serial independence of $\{\boldsymbol{\epsilon}(s)\}$ by Assumption (A3), we have $\widetilde{\boldsymbol{\Sigma}}_{-j}$, and hence $\mathbf{P}_{-j}$, are independent of $\widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)$ for each $j = 1, \ldots, L$. Writing $\mathbf{p}_{ji}$ as the ith eigenvector of $\mathbf{P}_{-j}$, we can then apply Lemma 1, 2 and 3 with $\mathbf{x}_i = \mathbf{p}_{ji}$ to get

$$\begin{aligned}
\max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \frac{\mathbf{p}_{ji}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{p}_{ji}}{\mathbf{p}_{ji}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{p}_{ji}} - 1 \right| &\leq \max_{\substack{i=1,\ldots,p \\ 1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}} \mathbf{B}_1 \mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)\mathbf{x}_i} - 1 \right| + \max_{\substack{i=1,\ldots,p \\ 1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}} \mathbf{B}_2 \mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)\mathbf{x}_i} \right| \\
&\quad + \max_{\substack{i=1,\ldots,p \\ 1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}} (\mathbf{B}_3 + \mathbf{B}_4) \mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)\mathbf{x}_i} \right| + \max_{\substack{i=1,\ldots,p \\ 1\leq \ell\leq L}} \left| \frac{\mathbf{x}_i^{\mathrm{T}} (\mathbf{B}_5 + \mathbf{B}_6) \mathbf{x}_i}{\mathbf{x}_i^{\mathrm{T}} \mathbf{\Sigma}(\tau_{\ell-1}, \tau_\ell)\mathbf{x}_i} \right| \\
&\overset{a.s.}{\to} 0.
\end{aligned}$$

Finally, note that

$$\begin{aligned}
&\max_{1\leq j\leq L} \left\| \widehat{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j) \mathbf{\Sigma}_{\mathrm{Ideal}}(\tau_{j-1}, \tau_j)^{-1} - \mathbf{I}_p \right\| \\
&= \max_{1\leq j\leq L} \left\| \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j}) \mathrm{diag}^{-1}(\mathbf{P}_{-j}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j}) - \mathbf{I}_p \right\| \\
&= \max_{\substack{i=1,\ldots,p \\ 1\leq j\leq L}} \left| \frac{\mathbf{p}_{ji}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{p}_{ji}}{\mathbf{p}_{ji}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{p}_{ji}} - 1 \right|,
\end{aligned}$$

which goes to 0 almost surely. This completes the proof of the theorem. □

Proof of Theorem 3. Define $\mathbf{D}_j = \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \mathbf{\Sigma}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$ and $\widetilde{\mathbf{D}}_j = \mathrm{diag}(\mathbf{P}_{-j}^{\mathrm{T}} \widetilde{\mathbf{\Sigma}}(\tau_{j-1}, \tau_j)\mathbf{P}_{-j})$. Then with $\mathbf{R} = \sum_{j=1}^{L} \mathbf{P}_{-j}(\widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p)\mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}}$,

$$\begin{aligned}
\widehat{\mathbf{\Sigma}}(0,1)^{-1} &= \left(\sum_{j=1}^{L} \mathbf{P}_{-j} \widetilde{\mathbf{D}}_j \mathbf{P}_{-j}^{\mathrm{T}} \right)^{-1} = \left(\sum_{j=1}^{L} \mathbf{P}_{-j}(\widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p)\mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}} + \sum_{j=1}^{L} \mathbf{P}_{-j} \mathbf{D}_j \mathbf{P}_{-j}^{\mathrm{T}} \right)^{-1} \\
&= (\mathbf{I}_p + \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1}\mathbf{R})^{-1} \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1} \\
&= \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1} + \sum_{k\geq 1} \big(- \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1}\mathbf{R} \big)^k \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1},
\end{aligned}$$

where the Neumann's series expansion in the last line is valid since

$$\begin{aligned}
\sum_{k\geq 0} \left\| \mathbf{\Sigma}_{\mathrm{Ideal}}(0,1)^{-1} \right\|^k \left\| \mathbf{R} \right\|^k &\leq 1 + \sum_{k\geq 1} \frac{\left\| \mathbf{R} \right\|^k}{\lambda_{\min}^k(\mathbf{\Sigma}_{\mathrm{Ideal}}(0,1))} \\
&\leq 1 + \sum_{k\geq 1} \frac{L^k \max_{1\leq j\leq L} \left\| \widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p \right\|^k \max_{1\leq j\leq L} \left\| \mathbf{\Sigma}(\tau_{j-1}, \tau_j) \right\|^k}{L^k \min_{1\leq j\leq L} \lambda_{\min}^k(\mathbf{\Sigma}(\tau_{j-1}, \tau_j))} \\
&\leq 1 + \sum_{k\geq 1} \left(\frac{C_2 C_4}{C_1 C_3} \max_{1\leq j\leq L} \left\| \widetilde{\mathbf{D}}_j \mathbf{D}_j^{-1} - \mathbf{I}_p \right\| \right)^k \\
&\overset{a.s.}{\to} 1 < \infty,
\end{aligned}$$

where C_1, C_2, C_3, C_4 are constants in Assumptions (A2) and (A4), and the last line follows from the asymptotic almost sure convergence of $\max_{1\le j\le L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p\big\|$ to 0 in Theorem 1. This implies that, asymptotically almost surely,

$$\begin{aligned}
\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\| &\le \lambda_{\max}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\sum_{k\ge1}\frac{\big\|\mathbf{R}\big\|^k}{\lambda_{\min}^k(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1))}\\
&\le \frac{1}{L\min_{1\le j\le L}\lambda_{\min}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}\cdot\sum_{k\ge1}\left(\frac{C_2C_4}{C_1C_3}\max_{1\le j\le L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p\big\|\right)^k\\
&\le \frac{1}{C_1C_3}\sum_{k\ge1}\left(\frac{C_2C_4}{C_1C_3}\max_{1\le j\le L}\big\|\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p\big\|\right)^k \overset{a.s.}{\to} 0. \qquad (7.14)
\end{aligned}$$

From the formulae of $\widehat{\mathbf{w}}_{\text{opt}}$ and $\mathbf{w}_{\text{Ideal}}$ given at the beginning of Section 3.1,

$$\text{Eff}(\mathbf{w}_{\text{Ideal}},\widehat{\mathbf{w}}_{\text{opt}})=\frac{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}\cdot\left(\frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}\right)^2.$$

With (7.14), the first term of the product on the right hand side above can be decomposed into $1+R_1+R_2$, where

$$\begin{aligned}
R_1&=\frac{\mathbf{1}_p^{\mathrm{T}}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}-\widehat{\boldsymbol{\Sigma}}(0,1)^{-1})\boldsymbol{\Sigma}(0,1)\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p},\\
R_2&=\frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}-\widehat{\boldsymbol{\Sigma}}(0,1)^{-1})\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p},
\end{aligned}$$

with, asymptotically,

$$\big\|R_1\big\|\le\frac{\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|\cdot\frac{C_2C_4}{C_1C_3}\lambda_{\max}^2(\widehat{\boldsymbol{\Sigma}}(0,1))}{C_1C_3}\overset{a.s.}{\to}0,$$

since we have (7.14) and that by Theorem 1, asymptotically almost surely, $\lambda^2_{\max}(\widehat{\boldsymbol{\Sigma}}(0,1))\le C_2C_4$. Similarly, we can prove that asymptotically, $R_2\overset{a.s.}{\to}0$, as well as

$$\frac{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}\overset{a.s.}{\to}1.$$

This completes the proof of the theorem. □

Proof of Theorem 4. Define $\mathbf{e}_i$ to be the unit vector with 1 on the ith position and 0 elsewhere, and $\big\|\mathbf{A}\big\|_1=\max_j\sum_i|a_{ij}|$ the L_1 norm of a matrix $\mathbf{A}$. For some $i=1,\dots,p$, using the same notations as in the proof of Theorem 3,

$$
\begin{aligned}
p^{1/2}\big\|\widehat{\mathbf{w}}_{\text{opt}}\big\|_{\max} &= \frac{p^{1/2}|\mathbf{e}_i^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p|}{\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p} \le \frac{p^{1/2}\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\big\|_1}{p\lambda_{\min}(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1})} \le \frac{p^{1/2}\cdot p^{1/2}/\lambda_{\min}(\widehat{\boldsymbol{\Sigma}}(0,1))}{p/\lambda_{\max}(\widehat{\boldsymbol{\Sigma}}(0,1))} \\
&\le \frac{\sum_{j=1}^{L}\lambda_{\max}(\widetilde{\mathbf{D}}_j)}{\sum_{j=1}^{L}\lambda_{\min}(\widetilde{\mathbf{D}}_j)} \\
&\le \frac{L\max_{1\le j\le L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)\lambda_{\max}(\mathbf{D}_j)+\sum_{j=1}^{L}\lambda_{\max}(\mathbf{D}_j)}{L\min_{1\le j\le L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)\lambda_{\min}(\mathbf{D}_j)+\sum_{j=1}^{L}\lambda_{\min}(\mathbf{D}_j)} \\
&\le \frac{C_2C_4\max_{1\le j\le L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)+\sum_{j=1}^{L}\lambda_{\max}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))}{C_1C_3\min_{1\le j\le L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)+\sum_{j=1}^{L}\lambda_{\min}(\boldsymbol{\Sigma}(\tau_{j-1},\tau_j))} \\
&= \frac{C_2C_4\max_{1\le j\le L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)+\lambda_{\max}\Big(\sum_{j=1}^{L}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\Big)}{C_1C_3\min_{1\le j\le L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)+\lambda_{\min}\Big(\sum_{j=1}^{L}\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)\Big)} \overset{a.s.}{\to} \mathrm{Cond}(\boldsymbol{\Sigma}(0,1)),
\end{aligned}
$$

where the last line follows from Assumption (A5) that each $\boldsymbol{\Sigma}(\tau_{j-1},\tau_j)$ has the same set of eigenvectors and the same order of eigenvalues, and that by Theorem 1 both

$$
\frac{\max_{1\le j\le L}\lambda_{\max}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)}{\lambda_{\min}(\boldsymbol{\Sigma}(0,1))}, \quad \frac{\min_{1\le j\le L}\lambda_{\min}(\widetilde{\mathbf{D}}_j\mathbf{D}_j^{-1}-\mathbf{I}_p)}{\lambda_{\min}(\boldsymbol{\Sigma}(0,1))}
$$

go to 0 asymptotically almost surely.

For the actual risk upper bound, consider the decomposition $pR(\widetilde{\mathbf{w}}_{\text{opt}})=I_1+I_2+I_3$, where

$$
\begin{aligned}
I_1 &= \frac{p\mathbf{1}_p^{\mathrm{T}}(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\boldsymbol{\Sigma}(0,1)\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}, \\
I_2 &= \frac{p\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)(\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}, \\
I_3 &= \frac{p\mathbf{1}_p^{\mathrm{T}}\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\boldsymbol{\Sigma}(0,1)\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\mathbf{1}_p}{(\mathbf{1}_p^{\mathrm{T}}\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}\mathbf{1}_p)^2}.
\end{aligned}
$$

By (7.14),

$$
|I_1| \le \frac{p^2\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|\cdot C_2C_4\cdot\big(\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|+\frac{1}{C_1C_3}\big)}{p^2\big(\lambda_{\min}(\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1})-\big\|\widehat{\boldsymbol{\Sigma}}(0,1)^{-1}-\boldsymbol{\Sigma}_{\text{Ideal}}(0,1)^{-1}\big\|\big)^2} \overset{a.s.}{\to} 0.
$$

Similarly, $|I_2| \stackrel{a.s.}{\to} 0$. For I_3, by (7.14),

$$
\begin{aligned}
|I_3| &\le \frac{p^2 \lambda_{\max}^2(\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1})\lambda_{\max}(\mathbf{\Sigma}(0,1))}{p^2\left(\lambda_{\min}(\mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}) - \left\|\widehat{\mathbf{\Sigma}}(0,1)^{-1} - \mathbf{\Sigma}_{\text{Ideal}}(0,1)^{-1}\right\|\right)^2} \\
&\stackrel{a.s.}{\to} \frac{\lambda_{\max}^2(\mathbf{\Sigma}_{\text{Ideal}}(0,1))}{\lambda_{\min}^2(\mathbf{\Sigma}_{\text{Ideal}}(0,1))}\lambda_{\max}(\mathbf{\Sigma}(0,1)) \\
&\le \left(\frac{\sum_{j=1}^{L} \lambda_{\max}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}{\sum_{j=1}^{L} \lambda_{\min}(\mathbf{\Sigma}(\tau_{j-1},\tau_j))}\right)^2 \lambda_{\max}(\mathbf{\Sigma}(0,1)) \\
&= \left(\frac{\lambda_{\max}(\sum_{j=1}^{L} \mathbf{\Sigma}(\tau_{j-1},\tau_j))}{\lambda_{\min}(\sum_{j=1}^{L} \mathbf{\Sigma}(\tau_{j-1},\tau_j))}\right)^2 \lambda_{\max}(\mathbf{\Sigma}(0,1)) = \text{Cond}^2(\mathbf{\Sigma}(0,1))\lambda_{\max}(\mathbf{\Sigma}(0,1)),
\end{aligned}
$$

where the last line follows from Assumption (A5). This completes the proof of the theorem. □

References

Abadir, K. M., Distaso, W., and Žikeš, F. (2014). Design-free estimation of variance matrices. *Journal of Econometrics*, 181(2):165 – 180.

Aït-Sahalia, Y., Fan, J., and Xiu, D. (2010). High-frequency covariance estimates with noisy and asynchronous financial data. *Journal of the American Statistical Association*, 105(492):1504–1517.

Aït-Sahalia, Y., Mykland, P. A., and Zhang, L. (2005). How often to sample a continuous-time process in the presence of market microstructure noise. *Review of Financial Studies*, 18(2):351–416.

Asparouhova, E., Bessembinder, H., and Kalcheva, I. (2013). Noisy prices and inference regarding returns. *The Journal of Finance*, 68(2):665–714.

Bai, Z. and Silverstein, J. (2010). *Spectral Analysis of Large Dimensional Random Matrices*. Springer Series in Statistics, New York, 2 edition.

Bai, Z. D. and Silverstein, J. (1998). No eigenvalues outside the support of the limiting spectral distribution of large-dimensional sample covariance matrices. *The Annals of Probability*, 26(1):316–345.

Barndorff-Nielsen, O. E., Hansen, P. R., Lunde, A., and Shephard, N. (2011). Multivariate realised kernels: Consistent positive semi-definite estimators of the covariation of equity prices with noise and non-synchronous trading. *Journal of Econometrics*, 162(2):149 – 169.

Fan, J., Li, Y., and Yu, K. (2012). Vast volatility matrix estimation using high- frequency data for portfolio selection. *Journal of the American Statistical Association*, 107(497):412–428.

Griffin, J. E. and Oomen, R. C. (2011). Covariance measurement in the presence of non-synchronous trading and market microstructure noise. *Journal of Econometrics*, 160(1):58 – 68. Realized Volatility.

Lam, C. (2016). Nonparametric eigenvalue-regularized precision or covariance matrix estimator. *Ann. Statist.* To appear.

Ledoit, O. and Wolf, M. (2012). Nonlinear shrinkage estimation of large-dimensional covariance matrices. *The Annals of Statistics*, 40(2):1024–1060.

Markowitz, H. (1952). Portfolio selection. *The Journal of Finance*, 7(1):77–91.

Tao, M., Wang, Y., Yao, Q., and Zou, J. (2011). Large volatility matrix inference via combining low-frequency and high-frequency approaches. *Journal of the American Statistical Association*, 106(495):1025–1040.

Wang, Y. and Zou, J. (2010). Vast volatility matrix estimation for high-frequency financial data. *Ann. Statist.*, 38(2):943–978.

Xiu, D. (2010). Quasi-maximum likelihood estimation of volatility with high frequency data. *Journal of Econometrics*, 159(1):235 – 250.

Zhang, L. (2011). Estimating covariation: Epps effect, microstructure noise. *Journal of Econometrics*, 160(1):33 – 47.